
STEERING THE HERD: A FRAMEWORK FOR LLM-BASED CONTROL OF SOCIAL LEARNING

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ABSTRACT

Algorithms increasingly govern access to information, from social media feeds and targeted advertising to LLM-based search interfaces, recommenders, and personal assistants. These systems create a coupled sequential decision environment: each agent receives a private algorithmically mediated signal while also learning from the publicly observed actions of earlier agents. We develop a model of controlled social learning in which an information-mediating planner, such as an LLM, chooses the precision of agents’ private signals while agents retain decentralized Bayesian control over their own actions. The planner may be altruistic, optimizing social welfare, or biased, seeking to induce a preferred action independently of the state. This formulation yields a dynamic optimization framework that links planner control, decentralized action choice, and Bayesian public-belief propagation. We prove convexity of the altruistic value function and characterize optimal altruistic and biased policies as functions of the evolving public belief. The resulting policies partition the belief space into interpretable regimes of maximal investment, non-intervention, and precision adjustment; in particular, a biased planner may optimally reduce signal informativeness to preserve a favorable public belief. Even under conservative transparency constraints—information parity with individuals, no lying or cherry-picking, and observable control choices—information mediation can substantially improve or degrade social welfare. We complement the theoretical analysis with LLM-based simulations in which LLMs instantiate both planner and agents. The LLM planner exhibits emergent strategic precision control that broadly follows the analytically predicted policy structure, while systematic departures from the Bayesian benchmark reflect non-Bayesian belief-updating patterns in the simulated agents. Together, the theory and simulations establish controlled social learning as a tractable framework for analyzing the welfare consequences and governance of LLM information mediators.

1 INTRODUCTION

Large Language Models (LLMs) now function as information mediators in socially consequential settings, including search engines and news aggregators (Zhang et al., 2024a; Gao et al., 2024) as well as personal assistants that supply medical or political advice (Huo et al., 2025; Haupt & Marks, 2023; Schiele et al., 2024; Argyle et al., 2023). Their informational value is accompanied by documented capacities for persuasion (Potter et al., 2024; Carrasco-Farre, 2024; Rogiers et al., 2024), manipulation (Williams et al., 2025; Liu et al., 2025; Jones & Bergen, 2024), and strategic interaction (Lorè & Heydari, 2024; Payne & Alloui-Cros, 2025; Zhang et al., 2024b). At scale, such influence is not exerted on isolated users alone. It operates in social environments where people observe one another, draw inferences from prior behavior, and update collectively. Algorithmic mediation can therefore reshape not only individual decisions, but also the social-learning process through which those decisions become evidence for subsequent agents.

A recommendation problem illustrates the mechanism. An LLM-powered system presents a new restaurant to a sequence of potential customers. The restaurant’s quality—the payoff-relevant state—is either good or bad, yet neither the users nor the system directly observes that state. Each user receives a personalized recommendation, represented as a private signal, and then decides whether to patronize the restaurant. The recommendation system, which we model as the *planner*, can invest resources to make the signal more informative, for example by producing a review more

carefully tailored to the user’s preferences. Signal precision thus summarizes the cost of market research, surveys, data acquisition, and other forms of personalization that improve information quality without revealing the state itself.

The user’s decision is not determined by the private signal alone. Users also observe the actions of their predecessors—if the previous ten users patronized the restaurant, the next user rationally infers that those users likely received favorable information. This process, known as *social learning*, shapes behavior in domains such as vaccine adoption (e.g., Rao et al. (2007); Bauch & Bhattacharyya (2012)), technology adoption (e.g., Gillingham & Bollinger (2021); Weber (2012)), and political and moral opinion formation (e.g., Brady et al. (2021); Guilbeault et al. (2018)). The central object of this paper is the strategic interaction between a planner that controls the precision of private information and a population that learns from public actions. Because each induced action becomes evidence for later agents, the planner’s current precision choice creates an information *externality*. Optimal mediation must therefore be analyzed as a dynamic control problem rather than as a sequence of independent recommendations.

We develop a tractable model of *controlled social learning*. An algorithmic planner sequentially chooses, at a cost, the precision of private signals delivered to agents who also observe predecessor actions. The planner’s control is deliberately constrained: it neither falsifies information, suppresses histories, nor commands actions, but instead determines how much to invest in making each signal informative. This formulation isolates three questions central to algorithmic information mediation: how precision control steers collective beliefs, how optimal precision responds to the evolving public belief, and how welfare changes when the planner is altruistic rather than biased.

The planner’s objective determines the normative content of the same informational instrument. An *altruistic* planner, analogous to an ideal recommendation system, maximizes user utility by helping agents choose the action that matches the true state. A *biased* planner, such as a recommender with a side payment from the restaurant, seeks instead to induce a fixed action regardless of the state. Precision control can therefore implement welfare-enhancing clarification or strategically motivated steering.

This distinction is related to alignment but does not collapse into it. The altruistic planner is aligned with agents’ *expected utility* under uncertainty. The biased planner has a state-independent objective and is therefore not aligned in that expected-utility sense, although its preferred action may coincide with agents’ *realized utility* when the true state favors that action. The framework consequently captures both beneficial and harmful instances of biased mediation without assuming that bias necessarily materializes as welfare loss.

Contributions and Outline The paper makes three principal contributions that connect the formal model, the policy analysis, and the LLM-based evaluation:

1. **A framework for controlled social learning.** We formulate a dynamic control problem in which a centralized information mediator selects the precision of private signals while agents retain decentralized control over their actions and update from observed predecessor behavior. The model accommodates both an altruistic objective that maximizes social welfare and a biased objective that seeks to induce a specified action.
2. **Optimal policy characterizations.** We characterize the optimal policies of altruistic and biased planners as functions of the public belief. For the altruistic planner, the characterization follows from a convexity theorem for the value function. For the biased planner, the analysis identifies belief regimes in which persuasion, obfuscation, and non-intervention are each optimal responses to the informational externalities generated by social learning.
3. **LLM-based empirical evaluation.** We simulate controlled social-learning environments in which LLMs act as both planner and agents. The experiments show that social learning materially changes planner behavior and welfare outcomes, that the LLM planner’s emergent strategy broadly follows the theoretically predicted structure, and that its deviations from the Bayesian benchmark are consistent with systematic non-Bayesian behavior by the LLM agents.

Section 2 positions the framework relative to social learning, information design, online persuasion, and LLM-based persuasion. Section 3 develops the controlled social-learning model; Sections 4 and

5 derive the optimal policy structures; Section 6 evaluates the theory through LLM-based simulations; and Section 7 concludes. The appendices provide proofs, extensions, prompts, and additional validation.

2 RELATED WORK

The framework connects four literatures while occupying a distinct position within each: social learning, information design, online persuasion and reinforcement learning, and recent work on LLMs as strategic information intermediaries.

Social Learning Sequential social learning studies how agents infer an unknown state from private information together with the observable actions of predecessors. Because each action partially reveals the acting agent’s information, individual decisions become public inputs to subsequent beliefs. The canonical economic models of this process were introduced by Bikhchandani et al. (1992) and Banerjee (1992), with later work characterizing social-learning dynamics in broader environments (e.g., Smith & Sørensen (2000); Arieli & Mueller-Frank (2021)). Related studies incorporate richer social information and behavioral distortions, including biases and misperceptions (see Dasaratha & He (2022); Bistriz et al. (2022); Eyster & Rabin (2010) as well as Bikhchandani et al. (2024) and the references therein).

Work on the *control* of social learning is more limited (Wei & Anastasopoulos, 2022; Smith et al., 2021; Krishnamurthy, 2012; Bhatt & Krishnamurthy, 2021). The closest points of comparison are Wei & Anastasopoulos (2022) and Smith et al. (2021). Wei & Anastasopoulos (2022) assumes two-way communication, allowing agents to report private signals to the planner in exchange for action recommendations. Smith et al. (2021) studies a benevolent planner who directly modifies agents’ choice rules and derives a decentralized pivot mechanism to improve learning. Our model instead keeps the planner informationally constrained, requires no private-signal reports, and preserves agents’ autonomy over actions. These restrictions make the framework directly suited to algorithmic information mediators that influence users through information quality rather than through commands, transfers, or explicit mechanism participation.

Information Design Because the planner optimizes by choosing private signal precision, the model is also a constrained information-design problem (Bergemann & Morris, 2019). The modern literature builds from Bayesian persuasion, in which a sender chooses the signal structure observed by a rational Bayesian receiver who is uncertain about the state (Kamenica & Gentzkow, 2011). In each period of our model, the planner faces a costly and restricted version of this design problem: the signal is binary and symmetric, and deviations in precision are costly. The key distinction is intertemporal. The receiver’s action becomes social information for future receivers, so the value of precision depends not only on the current action but also on the public-belief trajectory induced by that action.

Only a small set of papers combines information design with sequential social learning (Arieli et al., 2022; Wu et al., 2025). These works study one-shot design: the sender selects an information structure at the outset, and that structure remains fixed for the entire sequence. Arieli et al. (2022) allows a self-interested sender to choose a general Bayesian-persuasion signal structure, while Wu et al. (2025) restrict the structure to be more informative in the Blackwell sense and maximizes the probability of a correct cascade. By contrast, our sender chooses a new signal precision for each agent, making information design dynamic and tying the optimal signal choice directly to the current public belief.

Online Persuasion and RL A growing literature connects online learning with Bayesian persuasion and information design. These papers study online sequential persuasion under unknown environment parameters, where a sender repeatedly interacts with short-lived receivers without knowing the prior over states and/or receivers’ utility functions (e.g. Castiglioni et al. (2020); Bacchiocchi et al. (2024); Castiglioni et al. (2021); Agrawal et al. (2023)). A related line analyzes sequential persuasion in dynamic environments or MDPs, introducing state transitions and/or receiver foresight; these models emphasize history-dependent or promise-based signaling policies and often connect to reinforcement learning (e.g. Wu et al. (2022); Su et al. (2022); Bernasconi et al. (2023)). The central objects in this literature are typically learning-based performance guarantees, such as sublinear

regret relative to the best hindsight signaling scheme, or approximation algorithms under computational constraints.

Our model isolates a different source of sequential dependence. The planner need not learn unknown environment parameters and is not informationally advantaged: it observes neither the underlying state nor agents’ private signals. Instead, the state variable evolves because receivers learn socially from earlier actions. Although the planner’s problem is sequential, it is not an online-learning or RL environment in the usual sense: actions do not modify the payoff-relevant state, and the planner does not explore to estimate unknown environmental parameters. The planner updates only through observed actions, which encode social learning and belief propagation. This information externality is precisely what makes the optimal precision policy depend on the public belief.

LLMs Recent work has begun to examine LLMs in information design and persuasion (Harris et al., 2025; Cheng & You, 2025; Raifer et al., 2022; Bai et al., 2024; Duetting et al., 2025). These studies primarily analyze one-on-one interactions and therefore do not model the social feedback channel central to our setting. The closest methodological antecedent is Duetting et al. (2025), who study a sender that chooses both a signal structure and a framing that can shape the posterior belief of a possibly non-Bayesian receiver. They use an LLM-based oracle to search over the computationally difficult framing space, building on evidence that LLMs can approximate humans as economic agents (Dillion et al., 2023; Horton, 2023). Their model does not address repeated interaction or social learning; we adapt elements of their empirical methodology to evaluate LLM planners and agents in the controlled social-learning simulations of Section 6.

3 SOCIAL LEARNING MODEL

We study a planner who interacts with a countable sequence of Bayes-rational agents indexed by $i \in \mathbb{N}_{>0}$. At time $i = 0$, nature draws a fixed, *unknown* exogenous state $\omega \in \Omega := \{G, B\}$ with common prior $\mathbb{P}(\omega = G) = b_1$. In the recommendation example of Section 1, G denotes the state in which the restaurant is good.

At each time i , the planner selects the informativeness of the private signal delivered to agent i . The signal $s_i \in \Omega$ is binary and symmetric: it equals the realized state with probability $q_i \in [0.5, 1]$, so $\mathbb{P}(s_i = \omega) = q_i$. We refer to q_i as agent i ’s signal precision; larger values of q_i represent more reliable personalized information.

Conditional on ω , each agent’s signal is independent of all other agents’ signals and of the prior history. Agent i observes her signal and the actions of all predecessors $j < i$, then selects an action $a_i \in \Omega$. She receives utility 0 when $a_i = \omega$ and utility $-C$ otherwise, with $C > 0$, and chooses an action that maximizes expected utility. In the recommendation example, this decision corresponds to patronizing the business exactly when the state is G .

3.1 AGENTS’ DECISION PROBLEMS

Each agent observes all predecessor actions and their associated signal precisions. We denote this public history by $\mathcal{H}_i := (b_1, (q_j, a_j)_{j < i})$. Because the planner and all agents share $\mathcal{H}_i$, they also share a *public* belief about ω . Immediately before agent i acts, this belief is $b_i = \mathbb{P}(\omega = G | \mathcal{H}_i)$, with initial value b_1 . As in the classical models of Banerjee (1992) and Bikhchandani et al. (1992), the public-belief process $(b_i)_{i \in \mathbb{N}}$ is Markovian and b_i is a sufficient statistic for $\mathcal{H}_i$ (see Appendix B.3 for proof).

Given $\mathcal{H}_i$ (equivalently, by Markovianity, b_i) and q_i , agent i chooses $a_i \in \Omega$ to maximize expected utility, breaking indifference in favor of the action that matches s_i . Her posterior, or *private* belief, after observing s_i is (derivation in Appendix B.1):

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q_i}{1+2b_iq_i-b_iq_i} b_i & s_i = G \\ \frac{1-q_i}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (1)$$

The induced decision rule selects G when $\tilde{b}_i > 0.5$, selects B when $\tilde{b}_i < 0.5$, and follows the private signal when $\tilde{b}_i = 0.5$. Substituting the posterior expression yields the threshold representation

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases} \quad (2)$$

Subsequent agents observe a_i , and the public belief evolves according to the transition map below (see Equation (14), Appendix B.2):

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (3)$$

Remark 1. When $1 - q_i \leq b_i \leq q_i$, Equation (2) implies that agent i 's action fully reveals her private signal, so the updated public belief coincides with agent i 's private belief. Outside this interval, the public belief is sufficiently strong that the private signal cannot change the action. The action is then uninformative, the public belief remains fixed, and the process enters the absorbing state known in the social-learning literature as an information cascade or herding (Bikhchandani et al., 1992; Banerjee, 1992). At such states, the public belief no longer incorporates agents' private information.

Agent i 's expected utility is $-C \mathbb{P}(a_i \neq \omega | b_i, q_i)$, which reduces to the closed-form expression below (see Appendix C.1):

$$-C \mathbb{P}(a_i \neq \omega | b_i, q_i) = -C \min(b_i, 1 - b_i, 1 - q_i). \quad (4)$$

3.2 THE PLANNER'S PROBLEM

We analyze two planner objectives. The altruistic planner seeks to induce correct actions, $a_i = \omega$, whereas the biased planner seeks to induce a fixed action, taken without loss of generality to be G , independently of ω . We denote the corresponding objects by subscripts A and B . Under both objectives, the planner controls only the precision of each private signal. The cost of precision adjustment is represented by a function $\beta(\cdot)$ that is non-negative, increasing, continuous, and concave in its argument. Because the planner observes exactly the same public information $\mathcal{H}_i$ as the agents, influence operates through precision control rather than through privileged state information.

Altruistic Planner At time i , the altruistic planner chooses precision q_i and pays $\beta(q_i)$, with baseline precision $p \in [0.5, 1)$ normalized so that $\beta(p) = 0$. Because lowering precision cannot improve social welfare (Appendix C.9), the altruistic planner can be restricted without loss to precision improvements above p . Additional precision is costly, and concavity of $\beta(\cdot)$ implies weakly decreasing marginal cost. Agents know both p and $\beta(\cdot)$.

The altruistic planner maximizes discounted social welfare net of precision-investment costs, where social welfare is the agents' expected total utility. Let $r_A(b_i, q_i)$ denote the instantaneous reward when the public belief is b_i and the planner chooses precision q_i for agent i . Starting from b_1 and following policies $\pi = (\pi_i)_{i=1}^{\infty}$ with $\pi_i(\mathcal{H}_i) = q_i$, the planner obtains the following discounted value for discount factor $\delta \in [0, 1)$:

$$V_A^\pi(b_1) = \sum_{i=1}^{\infty} \delta^{i-1} r_A(b_i, \pi(b_i)), \quad \text{where} \quad r_A(b_i, q_i) = -\beta(q_i) - C \mathbb{P}(a_i \neq \omega | b_i, q_i). \quad (5)$$

The altruistic optimal value and policy are the supremum and arg supremum of $V_A^\pi(\cdot)$ over $\pi \in \Pi$, where Π denotes the admissible policy class. Setting $\delta = 0$ yields the myopic benchmark, which removes the informational effect of current actions on future agents.

Biased Planner The biased planner has a different objective and, correspondingly, a different reward and cost structure. Appendix A gives the motivating example of a political campaign. The biased planner seeks to induce action G from every agent, independently of ω . If the agent chooses G , the planner incurs cost 0; if the agent chooses B , the planner incurs cost $C > 0$, regardless of the

true state. Unlike the altruistic planner, the biased planner may benefit from either increasing or decreasing precision relative to the baseline p . Any deviation from p requires tailoring the information presented to the agent and therefore incurs cost $\beta(|q_i - p|)$, with $\beta(0) = 0$.

The biased planner's expected instantaneous reward at time i is therefore

$$r_B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = B | b_i, q_i) \quad (6)$$

Using Equation (6), the biased-planner quantities $V_B^\pi(\cdot)$, $V_B^*(\cdot)$, and $\pi_B^*(\cdot)$ are defined analogously to the altruistic quantities $V_A^\pi(\cdot)$, $V_A^*(\cdot)$, and $\pi_A^*(\cdot)$ in Equation (5).

Agents' utilities remain state-contingent: each agent is better off choosing the action that matches ω . Consequently, when $\omega = B$, a biased planner that successfully induces G lowers the agent's utility. The altruistic planner is aligned with agents' expected utility because it seeks $a_i = \omega$ in every state. The biased planner is aligned with realized utility only in the state $\omega = G$.

For both objectives, the planner's problem is an infinite-horizon discounted stationary Markov Decision Process with state $b_i \in [0, 1]$, control $q_i \in [0.5, 1]$, and transition function Equation (3) (Puterman, 1990). A unique optimal value function therefore exists (Kallenberg, 2011). We restrict attention to deterministic Markov policies, so $\Pi := \{\pi : [0, 1] \rightarrow [0.5, 1]\}$ and $\pi(b_i) = q_i$.

Remark 2. *The model imposes three deliberately conservative restrictions. (1) The planner observes the same history as the agents. This rules out settings with richer planner-side data, but captures environments in which all parties rely on the same public evidence, such as early findings or survey data. (2) The planner controls signal precision through a binary symmetric channel. This excludes cherry-picking, censorship, and falsification, while retaining settings in which the mediator improves noisy information without changing its underlying content. (3) The planner's control choices are fully observable. This omits covert framing policies, but captures transparent mediation regimes in which users can observe the supplied level of information quality.*

4 OPTIMAL ALTRUISTIC POLICIES

We begin with the myopic benchmark, which isolates the consequences of ignoring the future informational effects of current precision choices. In the model, myopia is obtained by setting the discount factor to $\delta = 0$, so the planner optimizes only the instantaneous reward and $V_A^\pi(b) = r_A(b, \pi(b))$. The optimal myopic policy $\pi_A^0(\cdot)$ is

$$\pi_A^0(b) \in \arg \sup_{q \in [0.5, 1]} r_A(b, q) \quad \forall b \in [0, 1] \quad (7)$$

This benchmark also has a decentralized interpretation: each agent chooses the precision of her own private signal, pays the associated cost, and maximizes her own expected utility net of that cost, without internalizing the effect of her action on later agents.

Theorem 1. *The optimal myopic altruistic policy π_A^0 is given as follows:*

$$\pi_A^0(b) = \begin{cases} 1 & b \in (t_M, 1 - t_M) \\ p & \text{o.w.} \end{cases} \quad \text{where } t_M = \begin{cases} \frac{\beta(1)}{C} & \beta(1) < C(1 - p) \\ 0.5 & \text{o.w.} \end{cases}$$

Proof in Appendix C.2.

The myopic optimum has a sharp threshold structure. When the public belief is already sufficiently informative, the planner uses the baseline precision p and pays no additional cost. When the public belief is sufficiently uncertain, the planner supplies a perfect signal of precision 1. The threshold depends only on the cost of a perfect signal, $\beta(1)$, relative to the cost of an incorrect action, C . If $\beta(1) \geq C$, perfect information is too costly to justify myopically; then $t_M = 0.5$ and the belief interval on which precision 1 is selected is empty.

The dynamic problem is governed by a central structural property of the altruistic value function.

Theorem 2. *$V_A^*(\cdot)$ is convex with respect to public belief.*

The proof of Theorem 2 (Appendix C.3) is technically substantive because agents' actions depend endogenously on the public belief. If actions were independent of the belief process, as in settings such as Nyarko (1994), expected utility would be linear in the belief state and convexity would

follow directly. Here, social-learning thresholds make convexity a substantive property of the controlled belief process rather than a standard implication.

Convexity is the key analytical step that makes the optimal altruistic policy fully characterizable.

Theorem 3. *There exist d_A, t_A such that $0 < d_A \leq t_A \leq t_M \leq 0.5$ and*

$$\pi_A^*(b) = \begin{cases} p & b \in [0, d_A) \cup (1 - d_A, 1] \\ 1 & b \in (t_A, 1 - t_A) \\ \max(b, 1 - b) & \text{o.w.} \end{cases}$$

Furthermore, if $t_M < 0.5$, then $d_A < t_M$. See C.5 for the proof and figure 2a for an example.

The optimal altruistic policy divides the belief space into three interpretable regimes. First, as in the myopic benchmark, the planner does not invest when the public belief is extreme. The dynamic planner, however, requires a stronger public belief before stopping investment, because it values the informational externality that current actions create for future agents.

Second, when the public belief is close to 0.5 and therefore weakly informative, the dynamic optimum selects precision 1 whenever the cost permits. By Equation (2), the agent's action then matches the true state with probability 1, causing the public belief to move immediately to either 0 or 1.

Third, in the intermediate regime, the planner chooses the minimum precision $\max(b, 1 - b)$ that makes the agent's action responsive to the private signal (see Equation (2)). Any lower precision leaves the action fully determined by the public belief and therefore informationally redundant. This regime identifies the lowest-cost intervention that keeps peer observation informative.

5 OPTIMAL BIASED POLICIES

We again begin with the myopic benchmark. As in Section 4, the myopic biased policy $\pi_B^0(\cdot)$ maximizes the instantaneous reward $r_B(b, q)$. We characterize $\pi_B^0(\cdot)$ while accounting for an important technical distinction: because the biased objective can generate discontinuities in the state-action value, an optimal policy need not exist at every belief state (Puterman, 1990, sec. 2.3.1). Where the supremum is not attained, we use ϵ -optimal policies, denoted with an ϵ superscript.

Theorem 4. *There exist $t_1, \dots, t_5 \in (0, p)$, $t_1 < 1 - p \leq t_2 \leq t_3 < 0.5 \leq t_4 \leq t_5 < p$ so that:*

- (A) *If $b \in [0, t_1] \cup (1 - p, t_2) \cup [t_3, t_4] \cup (p, 1]$, then $\pi_B^0(b) = p$.*
- (B) *If $b \in (t_1, 1 - p] \cup [t_2, t_3)$, then $\pi_B^0(b) = 1 - b$.*
- (C) *If $b \in (t_4, t_5)$, then $\pi_B^0(b) = 1$.*
- (D) *If $b \in [t_5, p]$, then $\pi_B^0(b)$ does not exist, and $\pi_B^{\epsilon, 0}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

The proof appears in Appendix C.6. This myopic structure provides the benchmark for the dynamic biased policy.

Theorem 5. *There exist $t_1, t_2 \in [0, p]$ with $t_1 \leq 1 - p \leq 0.5 < t_2 < p$ so that:*

- (A) *If $b \leq t_1$ or $b > p$, then $\pi_B^*(b) = p$.*
- (B) *If $b \in (t_1, 1 - p]$, then $\pi_B^*(b) \geq p$.*
- (C) *If $b \in (1 - p, 0.5)$, then $\pi_B^*(b) \geq 1 - b$.*
- (D) *If $b \in [0.5, t_2)$ and $\pi_B^*(b)$ exists, then $\pi_B^*(b) \geq b$.*
- (E) *If $b \in (t_2, p]$, then $\pi_B^*(b)$ does not exist, and $\pi_B^{\epsilon}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

Theorem 5, proved in Appendix C.8 and illustrated in figure 2a, identifies five belief regimes that explain how a biased planner uses precision control.

In regime (A), the public belief is strong enough that the cost of steering the system can exceed the planner's benefit, even when $b < 0.5$ is unfavorable to the planner. The baseline precision p lies

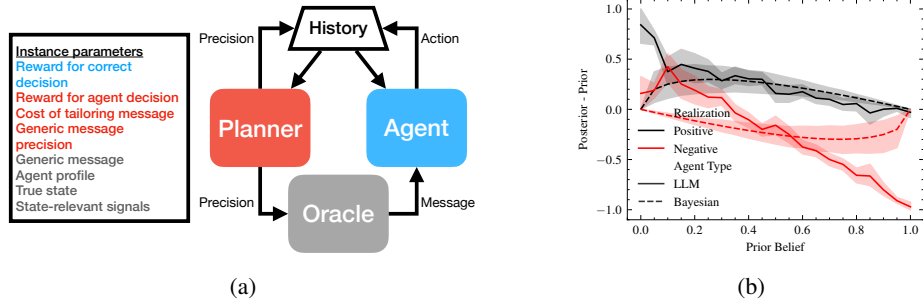


Figure 1: (a) System diagram for the LLM simulation. Instance parameters are color-coded by the LLM role that uses them. (b) Belief changes for LLM agents (solid) and Bayesian agents (dashed) after a positive (black) or negative (red) signal, conditional on the prior belief.

below $\max(b, 1 - b)$, so Equation (2) implies that the agent follows the public belief regardless of her private signal. The action is uninformative, the public belief is unchanged, and the Markovian policy repeats this behavior indefinitely. The resulting cascade is unfavorable to the planner when $b < 0.5$ and favorable otherwise.

In regime (B), the public belief is near an unfavorable cascade. The planner increases precision enough for the private signal to affect the agent’s action, even though the signal is expected to favor B . This investment preserves a positive probability of inducing the preferred action G and functions as a final opportunity to move the public belief away from an unfavorable absorbing region.

In regime (C), where the belief remains below 0.5 but is less extreme, the planner may reduce precision below p . Because more precise information is more likely to reveal unfavorable evidence, the planner chooses the least informative signal that still affects the agent’s action, namely a precision satisfying $q \geq 1 - b$. This preserves the possibility of improving the public belief while limiting the chance of strengthening the unfavorable belief.

In regime (D), the public belief weakly favors the planner’s desired action. Because $b > 0.5$, higher precision makes an agent more likely to infer $\omega = G$, so investment reinforces and increases the favorable public belief.

In regime (E), the public belief is sufficiently favorable that the planner optimally suppresses informative private variation by selecting precision just below $\max(b, 1 - b)$, namely $b - \epsilon$. Agents then ignore private signals and choose G . In this regime, the risk that private information could overturn a favorable public belief dominates both the cost of reducing precision and the possible benefit of further belief improvement.

6 EVALUATION VIA LLM-BASED SIMULATION

We use LLM-based simulation to evaluate the implications and applicability of the controlled social-learning framework. The evaluation proceeds in three steps: we first analyze the behavior of LLM agents to identify key deviations from Bayesian rationality, then examine how an LLM planner adapts its strategy in response, and finally evaluate the resulting impact on social welfare.

We instantiate the framework in an LLM-based simulation of controlled social learning in which agents decide whether to buy a new car. As shown in figure 1a, the simulation assigns LLMs to three roles, with the corresponding prompts reported in Appendix E.2.2:

Agent: Role-playing as an assigned profile, the agent observes prior actions and a private message about the car with known precision, reports a belief about quality, and chooses whether to buy.

Planner: Given the public action history, the planner selects the precision q_i for agent i ’s private signal according to its objective.

Oracle: Given the planner’s precision choice, an agent profile, and a car fact sheet (Appendix E.2.1), the oracle generates a tailored private signal with the requested precision. We compare the LLM simulations with numerical solutions of the planner’s MDP. In both settings, the cost function is linear, $\beta(q) = k|q - p|$, and we vary the slope k , baseline precision p , and discount factor δ , while

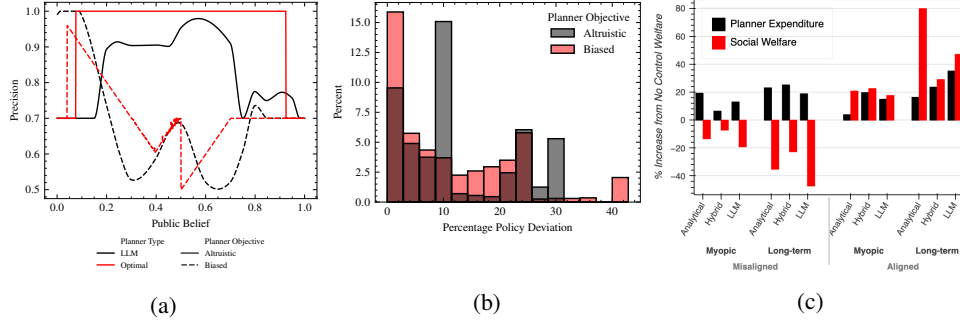


Figure 2: (a) Representative LLM-planner policies (black) and analytically optimal policies (red) for altruistic (solid) and biased (dashed) objectives. (b) Distribution of percentage deviations between LLM and optimal policies. (c) Planner expenditure and social-welfare change, each reported as a percentage of no-control baseline welfare. The true state is fixed to B; the left half shows a biased planner seeking action G, and the right half shows an altruistic planner.

fixing the cost of an incorrect action at $C = 1$. Appendix E gives the experimental setup and prompts, and Appendix E.3 validates the elicited beliefs and oracle precision.

The experiments support three conclusions:

1. **The impact of social-learning control:** Both analytical and LLM planners substantially improve their own objective values by exploiting social learning dynamics. The same control channel can improve or harm social welfare depending on whether planner and agent objectives are aligned.
2. **The robustness of the policy structure:** LLM planners choose policies that are structurally close to the analytically optimal policies even though the LLM agents are not Bayesian.
3. **The strategic behavior of LLM planners:** LLM planners exhibit emergent strategic behavior in their precision choices. Their policies not only approximate the theoretical structure but also deviate in directions that are consistent with the non-Bayesian behavior of the LLM agents they face.

6.1 LLMs AS NON-BAYESIAN AGENTS

While our analytical results assume Bayesian agents, both humans and LLMs exhibit systematic cognitive biases (Rupprecht et al., 2025). To isolate the relevant LLM deviations, we test how an LLM belief updates in response to new information. These results, shown in figure 1b, reveal three non-Bayesian patterns in LLM belief updating (see also Appendix E.4):

- (NB1) LLM agents underreact to private signals which align with their prior beliefs.
- (NB2) LLM agents overreact to private signals which run counter to their prior beliefs.
- (NB3) As a result, LLM agents require a stronger public belief to enter an information cascade.

These specific cognitive biases have also been observed empirically in human studies (e.g., Ba et al. (2022); Chan et al. (2025)). The subsequent results show that the LLM planner’s policy remains structurally aligned with the theoretical policy while adapting to non-Bayesian agents of this kind.

6.2 VALIDATION OF PLANNER POLICY STRUCTURE AND THE ROLE OF SOCIAL LEARNING

We now compare the policies of LLM planners with the optimal policies derived in Sections 4 and 5. As shown in figure 2a, despite the deviations in agent behavior, the emergent strategies of the LLM planner show clear structural similarity to the theoretical optimum. Thus, even when the optimal policy is highly non-trivial (e.g., the biased policy in figure 2a), the LLM planner exhibits strategic precision control that accounts for and capitalizes on social learning.

In the altruistic case, both planners cease investment when public belief is strong and invest heavily when it is uncertain. In the biased case, both planners exhibit the same qualitative trends: high investment to escape an unfavorable belief, reduced precision near the midpoint, and no investment once a favorable belief is established. The magnitude of the policy deviation is often modest, as shown in figure 2b. For both altruistic and biased planners, the deviation is less than 10% for the majority of belief states, underscoring the broad structural alignment between the emergent LLM strategy and the theoretical characterization.

The remaining structural differences are best understood as strategic adaptations to the specific non-Bayesian behaviors identified in Section 6.1. (1) The LLM planner tends to avoid extreme precisions (0.5 or 1.0), consistent with a known central tendency bias (Rupprecht et al., 2025). (2) The LLM planner’s policy tapers investment more gradually rather than imposing a sharp cutoff. This pattern responds to agents’ resistance to cascades (NB3): even agents with strong priors can be swayed by a countervailing private signal, so it is not entirely safe to stop investing. (3) Similarly, the biased LLM planner continues to invest at very low beliefs. This pattern is consistent with the possibility that agents overreact (NB2) to a surprisingly positive signal, making a “last-ditch effort” more viable than it would be against Bayesian agents.

6.3 WELFARE IMPLICATIONS OF ALTRUISM, BIAS, AND ALIGNMENT

We define social welfare $W^\pi(b)$ as the total expected discounted utility of all agents. Because better information never harms an agent’s expected utility, social welfare is monotonic in signal precision (see Appendix C.9). The altruistic planner therefore increases social welfare relative to the baseline. To quantify these effects, we compare social welfare and planner expenditure across three settings: (1) the *analytical* setting (optimal policy, Bayesian agents), (2) the *LLM* setting (LLM planner, LLM agents), and (3) a *hybrid* setting (optimal policy, LLM agents).

The results in figure 2c show that planners in all settings can substantially alter social welfare. Neglecting social learning, as in the myopic cases, also worsens outcomes for the planner. In particular, the biased analytical and LLM planners decreased social welfare by 40 to 50% when misaligned. They achieved this decrease by intentionally obscuring information about the true state through policies that reduce signal precision. The magnitude of the welfare loss is especially striking given the conservative restrictions imposed on the planners (Remark 2) and substantiates the risk posed by potentially misaligned LLM information mediators.

The results also highlight the cost of using a misspecified model of agents. In the hybrid setting, the analytically optimal policy, designed for Bayes-rational agents, is “brittle” and performs worse when applied to non-Bayesian agents. The LLM policy, by contrast, closely resembles the analytical policy while adjusting more effectively to non-Bayesian agents with human-like biases.

7 CONCLUSION

We introduced a framework for studying the strategic control of social learning by an algorithmic information mediator and developed a formal model of this increasingly important dynamic. The analysis shows that even a constrained planner can substantially affect social welfare, for better or for worse, and that social learning is central to the exercise of this influence. The simulations reinforce these theoretical conclusions: LLM planners exhibit strategic reasoning that reflects the structure of the analytical policies while adapting to the non-Bayesian behavior of LLM agents.

Taken together, the findings yield three implications for algorithmic information mediation. First, human-AI and human-algorithm interactions cannot be fully understood in isolation: their effects depend on the social-learning processes through which individual actions become evidence for others. Second, information mediators can guide or derail collective learning even when their formal control is limited to signal precision, which makes governance of this control channel consequential. Third, modern LLMs exhibit emergent strategic behavior that can account for social learning and for non-Bayesian cognitive biases. These implications underscore the need to understand how the risks of algorithmic information mediators, and LLMs in particular, can be mitigated. A limitation of the present study is the absence of human-subject data. Because the fidelity of LLM-human simulators remains contested (see, e.g., Horton (2023); Gao et al. (2025)), human data would be valuable for validating, and potentially calibrating, the LLM simulators.

Two future directions follow directly from the model. The first is to generalize the analysis beyond the maintained assumptions. Settings with larger state spaces, richer signal structures, more general loss functions, and heterogeneous agents would bring the framework closer to many empirical environments. Moving beyond binary states and 0-1 losses appears primarily algebraic, and we conjecture that the qualitative structure of the results will continue to hold. For more general signal structures, the principal challenge is preserving the convexity result of Theorem 2. Identifying the broadest class of signal structures for which such convexity holds is an important problem for future work. As a first step toward heterogeneity, Appendix D generalizes the altruistic planner results to agents with opposing preferences.

The second direction is to design safeguards against the welfare losses generated by biased planners. Natural candidates include regulations or mechanisms that better align planner incentives with agent welfare and thereby reduce detrimental effects from strategic information mediation.

REFERENCES

- Shipra Agrawal, Yiding Feng, and Wei Tang. Dynamic Pricing and Learning with Bayesian Persuasion. *Advances in Neural Information Processing Systems*, 36:59273–59285, December 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/b9c2e8a0bbcd5fcfaf62856a3a719ada-Abstract-Conference.html.
- Lisa P. Argyle, Christopher A. Bail, Ethan C. Busby, Joshua R. Gubler, Thomas Howe, Christopher Rytting, Taylor Sorensen, and David Wingate. Leveraging AI for democratic discourse: Chat interventions can improve online political conversations at scale. *Proceedings of the National Academy of Sciences*, 120(41):e2311627120, October 2023. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.2311627120. URL <https://pnas.org/doi/10.1073/pnas.2311627120>.
- Itai Arieli and Manuel Mueller-Frank. A general analysis of sequential social learning. *Mathematics of Operations Research*, 46:1235–1249, 11 2021.
- Itai Arieli, Ronen Gradwohl, and Rann Smorodinsky. Herd Design. In *Proceedings of the 23rd ACM Conference on Economics and Computation*, pp. 871–872, Boulder CO USA, July 2022. ACM. ISBN 978-1-4503-9150-4. doi: 10.1145/3490486.3538292. URL <https://dl.acm.org/doi/10.1145/3490486.3538292>.
- Cuimin Ba, J. Aislinn Bohren, and Alex Imas. Over- and Underreaction to Information. *SSRN Electronic Journal*, 2022. ISSN 1556-5068. doi: 10.2139/ssrn.4274617. URL <https://www.ssrn.com/abstract=4274617>. Publisher: Elsevier BV.
- Francesco Bacchiocchi, Matteo Bollini, Matteo Castiglioni, Alberto Marchesi, and Nicola Gatti. Online Bayesian Persuasion Without a Clue. *Advances in Neural Information Processing Systems*, 37:76404–76449, December 2024.
- Fengshuo Bai, Mingzhi Wang, Zhaowei Zhang, Boyuan Chen, Yinda Xu, Ying Wen, and Yaodong Yang. Efficient Model-agnostic Alignment via Bayesian Persuasion, May 2024.
- Abhijit V Banerjee. A simple model of herd behavior. *The Quarterly Journal of Economics*, 107(3):797–817, 1992.
- Larry M. Bartels. *Presidential primaries and the dynamics of public choice*. Princeton, N.J. : Princeton University Press, 1988.
- Chris T Bauch and Samit Bhattacharyya. Evolutionary game theory and social learning can determine how vaccine scares unfold. *PLoS computational biology*, 8(4), 2012.
- Dirk Bergemann and Stephen Morris. Information Design: A Unified Perspective. *Journal of Economic Literature*, 57(1):44–95, March 2019. ISSN 0022-0515. doi: 10.1257/jel.20181489. URL <https://www.aeaweb.org/articles?id=10.1257/jel.20181489>.
- Martino Bernasconi, Matteo Castiglioni, Alberto Marchesi, and Mirco Mutti. Persuading Farsighted Receivers in MDPs: The Power of Honesty. *Advances in Neural Information Processing Systems*, 36:14987–15014, December 2023.
- Sujay Bhatt and Vikram Krishnamurthy. Controlled Sequential Information Fusion With Social Sensors. *IEEE Transactions on Automatic Control*, 66(12):5893–5908, December 2021.

594 Sushil Bikhchandani, David Hirshleifer, and Ivo Welch. A theory of fads, fashion, custom, and cultural change
595 as informational cascades. *Journal of Political Economy*, 100(5):992–1026, 1992.

596 Sushil Bikhchandani, David Hirshleifer, Omer Tamuz, and Ivo Welch. Information Cascades and Social Learn-
597 ing. *Journal of Economic Literature*, 62(3):1040–1093, September 2024. ISSN 0022-0515, 2328-8175. doi:
598 10.1257/jel.20241472. URL <https://pubs.aeaweb.org/doi/10.1257/jel.20241472>.

599 Ilai Bistriz, Nasimeh Heydaribeni, and Achilleas Anastasopoulos. Informational cascades with nonmyopic
600 agents. *IEEE Transactions on Automatic Control*, 67, 9 2022.

601 William J Brady, Killian McLoughlin, Tuan N Doan, and Molly J Crockett. How social learning amplifies
602 moral outrage expression in online social networks. *Science Advances*, 7(33), 2021.

603 Carlos Carrasco-Farre. Large Language Models are as persuasive as humans, but how? About the cognitive
604 effort and moral-emotional language of LLM arguments, April 2024. URL [http://arxiv.org/abs/](http://arxiv.org/abs/2404.09329)
605 2404.09329. arXiv:2404.09329 [cs].

606 Matteo Castiglioni, Andrea Celli, Alberto Marchesi, and Nicola Gatti. Online Bayesian Persua-
607 sion. In *Advances in Neural Information Processing Systems*, volume 33, pp. 16188–16198. Cur-
608 ran Associates, Inc., 2020. URL [https://proceedings.neurips.cc/paper/2020/hash/](https://proceedings.neurips.cc/paper/2020/hash/ba5451d3c91a0f982f103cdbc249bc78-Abstract.html)
609 ba5451d3c91a0f982f103cdbc249bc78-Abstract.html.

610 Matteo Castiglioni, Alberto Marchesi, Andrea Celli, and Nicola Gatti. Multi-Receiver Online Bayesian Per-
611 suasion. In *Proceedings of the 38th International Conference on Machine Learning*, pp. 1314–1323. PMLR,
612 July 2021.

613 Kenneth Chan, Gary Charness, Chetan Dave, and J. Lucas Reddinger. On Prior Confidence and Belief Updat-
614 ing, May 2025. URL <http://arxiv.org/abs/2412.10662>. arXiv:2412.10662 [econ].

615 Zirui Cheng and Jiaxuan You. Towards Strategic Persuasion with Language Models, September 2025.

616 Krishna Dasaratha and Kevin He. Learning from viral content. *arXiv preprint arXiv:2210.01267*, 2022.

617 Danica Dillion, Niket Tandon, Yuling Gu, and Kurt Gray. Can AI language models replace human participants?
618 *Trends in Cognitive Sciences*, 27(7):597–600, July 2023. ISSN 1364-6613, 1879-307X. doi: 10.1016/j.
619 tics.2023.04.008. URL [https://www.cell.com/trends/cognitive-sciences/abstract/](https://www.cell.com/trends/cognitive-sciences/abstract/S1364-6613(23)00098-0)
620 S1364-6613(23)00098-0. Publisher: Elsevier.

621 Paul Duetting, Safwan Hossain, Tao Lin, Renato Paes Leme, Sai Srivatsa Ravindranath, Haifeng Xu, and Song
622 Zuo. Information Design With Large Language Models, September 2025. URL [http://arxiv.org/](http://arxiv.org/abs/2509.25565)
623 abs/2509.25565. arXiv:2509.25565 [cs].

624 Maxim Egorov, Zachary N. Sunberg, Edward Balaban, Tim A. Wheeler, Jayesh K. Gupta, and Mykel J.
625 Kochenderfer. POMDPs.jl: A framework for sequential decision making under uncertainty. *Journal of*
626 *Machine Learning Research*, 18(26):1–5, 2017.

627 Erik Eyster and Matthew Rabin. Naïve herding in rich-information settings. *American Economic Journal:*
628 *Microeconomics*, 2:221–243, 11 2010.

629 Shen Gao, Jiabao Fang, Quan Tu, Zhitao Yao, Zhumin Chen, Pengjie Ren, and Zhaochun Ren. Generative
630 News Recommendation. In *Proceedings of the ACM Web Conference 2024*, WWW ’24, pp. 3444–3453,
631 New York, NY, USA, May 2024. Association for Computing Machinery. ISBN 979-8-4007-0171-9. doi:
632 10.1145/3589334.3645448. URL <https://dl.acm.org/doi/10.1145/3589334.3645448>.

633 Yuan Gao, Dokyun Lee, Gordon Burtch, and Sina Fazelpour. Take Caution in Using LLMs as Hu-
634 man Surrogates: Scylla Ex Machina, January 2025. URL <http://arxiv.org/abs/2410.19599>.
635 arXiv:2410.19599 [econ].

636 Kenneth T Gillingham and Bryan Bollinger. Social learning and solar photovoltaic adoption. *Management*
637 *Science*, 67(11):7091–7112, 2021.

638 Douglas Guilbeault, Joshua Becker, and Damon Centola. Social learning and partisan bias in the interpretation
639 of climate trends. *Proceedings of the National Academy of Sciences*, 115(39):9714–9719, 2018.

640 Keegan Harris, Nicole Immorlica, Brendan Lucier, and Aleksandrs Slivkins. Algorithmic Persuasion Through
641 Simulation, February 2025. URL <http://arxiv.org/abs/2311.18138>. arXiv:2311.18138 [cs].

642 Claudia E. Haupt and Mason Marks. AI-Generated Medical Advice—GPT and Beyond. *JAMA*, 329(16):
643 1349–1350, April 2023. ISSN 0098-7484. doi: 10.1001/jama.2023.5321. URL [https://doi.org/](https://doi.org/10.1001/jama.2023.5321)
644 10.1001/jama.2023.5321.

-
- K. Hinderer. Lipschitz Continuity of Value Functions in Markovian Decision Processes. *Mathematical Methods of Operations Research*, 62(1):3–22, September 2005.
- John J. Horton. Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?, April 2023. URL <https://www.nber.org/papers/w31122>.
- Bright Huo, Amy Boyle, Nana Marfo, Wimonchat Tangamornsuksan, Jeremy P. Steen, Tyler McKechnie, Yung Lee, Julio Mayol, Stavros A. Antoniou, Arun James Thirunavukarasu, Stephanie Sanger, Karim Ramji, and Gordon Guyatt. Large Language Models for Chatbot Health Advice Studies: A Systematic Review. *JAMA Network Open*, 8(2):e2457879, February 2025. ISSN 2574-3805. doi: 10.1001/jamanetworkopen.2024.57879. URL <https://doi.org/10.1001/jamanetworkopen.2024.57879>.
- Cameron R. Jones and Benjamin K. Bergen. Lies, Damned Lies, and Distributional Language Statistics: Persuasion and Deception with Large Language Models, December 2024. URL <http://arxiv.org/abs/2412.17128>. arXiv:2412.17128 [cs].
- Lodewijk Kallenberg. Markov decision processes. *Lecture Notes. University of Leiden*, 428, 2011.
- Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6):2590–2615, 2011.
- Vikram Krishnamurthy. Quickest Detection POMDPs With Social Learning: Interaction of Local and Global Decision Makers. *IEEE Transactions on Information Theory*, 58(8):5563–5587, August 2012.
- LG AI Research. Exaone deep: Reasoning enhanced language models. *arXiv preprint arXiv:2503.12524*, 2025.
- Minqian Liu, Zhiyang Xu, Xinyi Zhang, Heajun An, Sarvech Qadir, Qi Zhang, Pamela J. Wisniewski, Jin-Hee Cho, Sang Won Lee, Ruoxi Jia, and Lifu Huang. LLM Can be a Dangerous Persuader: Empirical Study of Persuasion Safety in Large Language Models, April 2025. URL <http://arxiv.org/abs/2504.10430>. arXiv:2504.10430 [cs].
- Nunzio Lorè and Babak Heydari. Strategic behavior of large language models and the role of game structure versus contextual framing. *Scientific Reports*, 14(1), August 2024. ISSN 2045-2322. doi: 10.1038/s41598-024-69032-z. URL <https://www.nature.com/articles/s41598-024-69032-z>. Publisher: Springer Science and Business Media LLC.
- Yaw Nyarko. On the convexity of the value function in Bayesian optimal control problems. *Economic Theory*, 4(2):303–309, March 1994.
- Kenneth Payne and Baptiste Alloui-Cros. Strategic Intelligence in Large Language Models: Evidence from evolutionary Game Theory, July 2025. URL <http://arxiv.org/abs/2507.02618>. arXiv:2507.02618 [cs].
- Yujin Potter, Shiyang Lai, Junsol Kim, James Evans, and Dawn Song. Hidden Persuaders: LLMs’ Political Leaning and Their Influence on Voters, November 2024.
- Martin L. Puterman. Chapter 8 markov decision processes. In *Stochastic Models*, volume 2 of *Handbooks in Operations Research and Management Science*. Elsevier, 1990.
- Maya Raifer, Guy Rotman, Reut Apel, Moshe Tennenholtz, and Roi Reichart. Designing an Automatic Agent for Repeated Language-based Persuasion Games. *Transactions of the Association for Computational Linguistics*, 10:307–324, March 2022.
- Neel Rao, Markus M Mobius, and Tanya Rosenblat. Social networks and vaccination decisions. *FRB of Boston*, 2007.
- Alexander Rogiers, Sander Noels, Maarten Buyl, and Tijl De Bie. Persuasion with Large Language Models: a Survey, November 2024. URL <http://arxiv.org/abs/2411.06837>. arXiv:2411.06837 [cs].
- Jens Rupperecht, Georg Ahnert, and Markus Strohmaier. Prompt Perturbations Reveal Human-Like Biases in LLM Survey Responses, July 2025. URL <http://arxiv.org/abs/2507.07188>. arXiv:2507.07188 [cs].
- Dr.-Ing. Martin Schiele, Yannick Gittmann, Stefan Ilchmann, Ante Gojsalić, Dominik Jurinčić, and Phylis Klempt. Voting Advice Applications: Implementation of RAG-supported LLMs, July 2024. URL <https://www.techrxiv.org/users/802685/articles/1187651-voting-advice-applications-implementation-of-rag-supported-llms?commit=c540a54749906f8ac761093f5dff0f514291d4b6>.

-
- James E. Smith and Kevin F. McCardle. Structural Properties of Stochastic Dynamic Programs. *Operations Research*, 50(5):796–809, October 2002.
- Lones Smith and Peter Sørensen. Pathological outcomes of observational learning. *Econometrica*, 68:371–398, 2000.
- Lones Smith, Peter Norman Sørensen, and Jianrong Tian. Informational Herding, Optimal Experimentation, and Contrarianism. *The Review of Economic Studies*, 88(5):2527–2554, September 2021. ISSN 0034-6527, 1467-937X. doi: 10.1093/restud/rdab001. URL <https://academic.oup.com/restud/article/88/5/2527/6149316>.
- Shih-Tang Su, Vijay G. Subramanian, and Grant Schoenebeck. Bayesian Persuasion in Sequential Trials. In Michal Feldman, Hu Fu, and Inbal Talgam-Cohen (eds.), *Web and Internet Economics*, pp. 22–40, Cham, 2022. Springer International Publishing.
- Jeremy G Weber. Social learning and technology adoption: The case of coffee pruning in peru. *Agricultural Economics*, 43:73–84, 2012.
- Xupeng Wei and Achilleas Anastasopoulos. Social learning with a self-interested coordinator. In *2022 IEEE 61st Conference on Decision and Control (CDC)*, pp. 3137–3144. IEEE, 2022.
- Marcus Williams, Micah Carroll, Adhyyan Narang, Constantin Weisser, Brendan Murphy, and Anca Dragan. On Targeted Manipulation and Deception when Optimizing LLMs for User Feedback, February 2025. URL <http://arxiv.org/abs/2411.02306>. arXiv:2411.02306 [cs].
- Jibang Wu, Zixuan Zhang, Zhe Feng, Zhaoran Wang, Zhuoran Yang, Michael I. Jordan, and Haifeng Xu. Sequential Information Design: Markov Persuasion Process and Its Efficient Reinforcement Learning, February 2022.
- Shuo Wu, Pawan Poojary, and Randall Berry. Observational Learning with a Budget, April 2025. URL <http://arxiv.org/abs/2504.19396>. arXiv:2504.19396 [cs].
- Tianyi Zhang, Faisal Ladhak, Esin Durmus, Percy Liang, Kathleen McKeown, and Tatsunori B. Hashimoto. Benchmarking Large Language Models for News Summarization. *Transactions of the Association for Computational Linguistics*, 12:39–57, January 2024a. ISSN 2307-387X. doi: 10.1162/tacl_a_00632. URL https://doi.org/10.1162/tacl_a_00632.
- Yadong Zhang, Shaoguang Mao, Tao Ge, Xun Wang, Adrian de Wynter, Yan Xia, Wenshan Wu, Ting Song, Man Lan, and Furu Wei. LLM as a Mastermind: A Survey of Strategic Reasoning with Large Language Models, April 2024b. URL <http://arxiv.org/abs/2404.01230>. arXiv:2404.01230 [cs].

A APPLICATIONS ELABORATED

Consider a sequence of individuals (agents in our terminology) deciding whether to patronize a business or service provider (e.g., a restaurant, contractor, realtor, etc.) and a recommendation system (the planner) that provides information about the business. The true state of the world, assumed to be binary, is whether the business is good, and this state is unknown. Both the recommendation system and the agents seek actions consistent with the true state: agents want to patronize the business if and only if it is good, and the recommendation system wants to induce that same decision rule. In this setting, the recommendation system is an *altruistic planner*.

The recommendation system can show an agent a highly targeted (*precise* in our terminology) ad that presents the strengths and weaknesses of the business in contexts tailored to her background and characteristics, or it can show a generic or confusing ad that provides little decision-relevant information. A precise signal is more informative, comprehensible, and relatable for the agent, and is therefore more likely to lead her to the correct conclusion: patronize the business if and only if it is good. Social welfare increases as more agents make correct decisions. Precision adjustment is costly, however, because it requires tailoring content to the agent’s specific background. The recommendation system therefore chooses signal precisions to maximize social welfare net of tailoring costs.

As an example of a *biased planner*, consider voters (agents) who are motivated to support the candidate most likely to win in their community. The motivation to “back a winner” has been shown to influence electoral outcomes; for example, it helps explain why US states with early primary elections have an outsized impact on election results (see Bartels (1988)). The planner is a specific candidate’s campaign and therefore seeks to motivate agents to back that candidate. The binary true state indicates whether the candidate is winning. If an agent knew the true state, she would back the candidate if and only if he were winning.

Each agent understands her community well and would know whether the candidate is winning if she correctly understood his stances and policies. That understanding depends on receiving information in a form she can process effectively. Some agents may respond best to audio-visual material, others to long-form articles supported by facts, figures, and citations, others to brief and focused content, and others to material in their native language.

The campaign sends digital ads (private signals) to agents at varying degrees of precision. Precision represents the extent to which the content is tailored to the agent’s mode of understanding. A precise signal accurately conveys the true state by helping the agent understand the candidate’s policies, track record, and character, thereby enabling her to infer whether the candidate will win. Precision does not necessarily favor the campaign: if the candidate is losing in the agent’s community, a highly precise signal would lead the agent to oppose him. An imprecise signal, by contrast, may make backing the candidate more likely because it obfuscates the true state, for example by being vague or confusing, and thereby increases the chance that the agent believes he is winning. A biased planner may therefore have an incentive to decrease signal precision when it expects unfavorable information. Tailoring either precise or imprecise content requires research into how an agent processes information, so both increasing and decreasing precision are costly. The planner chooses precisions to maximize the expected number of backers net of the costs of generating those precisions.

B SOCIAL LEARNING DYNAMICS

We first summarize the sequence of events at step i :

1. The planner determines q_i based on $\mathcal{H}_i$, and q_i is observed by all agents.
2. Agent i receives private signal s_i such that $\mathbb{P}\{s_i = \omega\} = q_i$.
3. Agent i takes action a_i based on the observable history and her private signal, with the aim of maximizing her own utility.
4. All other agents and the planner observe a_i , and the public belief at the end of step i , b_{i+1} , is updated accordingly.

Because q_i is a function of $\mathcal{H}_i$, conditioning on both q_i and $\mathcal{H}_i$ makes the conditioning on q_i redundant. We use this principle throughout.

B.1 PRIVATE BELIEF UPDATE

We now derive agent i ’s action rule. Let $\alpha_i := \frac{b_i}{1-b_i}$ be the public likelihood ratio based on observed actions. Agent i forms the private belief $\tilde{b}_i = \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i\}$ by combining the current public belief with her private signal s_i . We define the corresponding private likelihood ratio as $\tilde{\alpha}_i := \frac{\tilde{b}_i}{1-\tilde{b}_i}$ and compute $\tilde{\alpha}_i$ and $\tilde{b}_i$ below.

1. If private signal is $s_i = G$, then

$$\begin{aligned} \tilde{\alpha}_i &= \frac{\tilde{b}_i}{1-\tilde{b}_i} = \frac{\mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i = G\}}{\mathbb{P}\{\omega = B | \mathcal{H}_i, q_i, s_i = G\}} \\ &= \frac{\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i\}}{\mathbb{P}\{s_i = G | \omega = B, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = B | \mathcal{H}_i, q_i\}} \\ &= \frac{q_i}{1-q_i} \frac{b_i}{1-b_i} \end{aligned} \tag{8}$$

$$= \frac{q_i}{1-q_i} \alpha_i \tag{9}$$

In obtaining Equation (8), we use the conditional independence of agents’ signals given ω to assert that $\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i\} = \mathbb{P}\{s_i = G | \omega = G\} = q_i$. The same argument applies when conditioning on $\omega = B$. We also use the fact that q_i is a function of $\mathcal{H}_i$.

2. If private signal is $s_i = B$, following the logic above,

$$\tilde{\alpha}_i = \frac{1-q_i}{q_i} \alpha_i \tag{10}$$

Let $y(b, q)$ and $z(b, q)$ denote the probabilities of realizing signals G and B , respectively, given public belief b and signal precision q . Conditioning on ω and applying the law of total probability gives

$$\begin{aligned} y(b_i, q_i) &= \mathbb{P}\{s_i = G | \mathcal{H}_i\} \\ &= \sum_{k \in \{G, B\}} \mathbb{P}\{s_i = G | \mathcal{H}_i, \omega = k\} \mathbb{P}\{\omega = k | \mathcal{H}_i\} \\ &= q_i b_i + (1 - q_i)(1 - b_i) = 1 + 2q_i b_i - q_i - b_i \end{aligned} \quad (11)$$

Similarly,

$$z(b_i, q_i) = q_i + b_i - 2q_i b_i \quad (12)$$

Combining Equation (9), Equation (10), Equation (11), and Equation (12) yields the following:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = f(b_i, q_i, s_i) = \begin{cases} \frac{q_i}{y(b_i, q_i)} b_i & s_i = G \\ \frac{1 - q_i}{z(b_i, q_i)} b_i & s_i = B \end{cases} \quad (13)$$

If $s_i = G$, then $y(b_i, q_i) = \mathbb{P}(s_i = G | \mathcal{H}_i)$ cannot be 0; similarly, if $s_i = B$, then $z(b_i, q_i)$ cannot be 0. The expression above is therefore well-defined on the realized signal path.

B.2 AGENT ACTIONS AND PUBLIC BELIEF

Agent i chooses G if $\tilde{b}_i$ exceeds 0.5 (equivalently, if $\tilde{\alpha}_i > 1$), chooses B if $\tilde{b}_i < 0.5$ (equivalently, if $\tilde{\alpha}_i < 1$), and follows s_i in the tie case. This action and i 's precision q_i are observed by all peers and determine the next public belief.

When $b_i < 1 - q_i$, then $\alpha_i < \frac{1 - q_i}{q_i}$. Therefore, if $s_i = G$, $\tilde{\alpha}_i < 1$ by Equation (9); if $s_i = B$, $\tilde{\alpha}_i < \frac{(1 - q_i)^2}{q_i^2} \leq 1$ (since $q_i \geq 0.5$) by Equation (10). Thus, regardless of the private signal realization s_i , $a_i = B$ when $b_i < 1 - q_i$. A symmetric argument shows that when $b_i > q_i$, $a_i = G$. These are precisely the cases in which the public belief is strong enough to dominate the private signal.

Now consider the case that $1 - q_i \leq b_i \leq q_i$. Then if $s_i = G$, $\tilde{\alpha}_i \geq 1$ from Equation (8) and since $b_i \geq 1 - q_i$. If $s_i = B$, then $\tilde{\alpha}_i \leq 1$ from Equation (10) and since $b_i \leq q_i$. Together with the tie-breaking rule, this results in action $a_i = s_i$. This leads to Equation (2), reproduced below.

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases}$$

Thus, agent i 's action is informative only in the first case of Equation (2), where it allows future agents to perfectly infer i 's private signal. In that case, the public belief updates as the private belief in Equation (13). In the latter two cases of Equation (2), the action contains no additional information about the true state, so the public belief remains unchanged. This yields Equation (3), reproduced below:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (14)$$

B.3 MARKOVIANITY AND SUFFICIENCY OF PUBLIC BELIEF

We now prove that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ depends on the public history only through b_i and q_i , for each $a \in \{G, B\}$. It follows that

$$\mathbb{P}(a_i = a | \mathcal{H}_i, q_i) = \mathbb{P}(a_i = a | b_i, q_i) \quad (15)$$

From Equation (2), the second and third cases immediately imply that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ is a function of b_i and q_i for $a \in \{G, B\}$, independent of the remaining components of $\mathcal{H}_i$. We consider the first case next.

$$\begin{aligned} \mathbb{P}(a_i = G | \mathcal{H}_i, q_i) &= \sum_{w \in \{G, B\}} \mathbb{P}(\omega = w | \mathcal{H}_i, q_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = w) \\ &= b_i \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = B) \\ &= b_i \mathbb{P}(s_i = G | b_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(s_i = G | b_i, q_i, \omega = B) \\ &= b_i q_i + (1 - b_i)(1 - q_i) \end{aligned} \quad (16)$$

The first step applies the law of total probability. In the second step, using the fact that q_i is a function of $\mathcal{H}_i$, we substitute b_i for $\mathbb{P}(\omega = w|\mathcal{H}_i, q_i)$. In the third step, we use the fact that, for $1 - q_i \leq b_i \leq q_i$, $a_i = s_i$, and, further, that s_i is independent of the history when conditioned upon ω . The result follows.

This leads to the following theorem:

Theorem 6. *Markovianity of the Public Belief Process*

$$b_{i+1} = \mathbb{P}\{\omega = G|b_i, q_i, a_i\} \quad (17)$$

In other words, b_i captures the entire information pertinent to the update of b_{i+1} that is contained in $\mathcal{H}_i$.

Proof. Note that, from step (1) in Appendix B, q_i is a function of $\mathcal{H}_i$. It suffices to show

$$b_{i+1} = \frac{\mathbb{P}\{a_i \cap \omega = G|b_i, q_i\}}{\mathbb{P}\{a_i|b_i, q_i\}}$$

We will prove the claim via induction on i .

For our base case, consider $i = 1$ so that $\mathcal{H}_2 = \{b_1, (q_1, a_1)\}$ and $b_i = b_1$. An application of Bayes' Theorem provides the following:

$$\begin{aligned} b_2 &= \frac{\mathbb{P}\{\omega = G|b_1, q_1\} \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} = \frac{b_1 \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \\ &= \frac{b_1 \mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\} \mathbb{P}\{\omega = G|b_1, q_1\}} \\ &= \frac{\mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \end{aligned}$$

where the final step follows as the unconditioned belief of ω is equal to the prior belief b_1 .

Now, assume the claim holds for all $i \leq n$.

$$\begin{aligned} b_{n+1} &= \mathbb{P}\{\omega = G|\mathcal{H}_n, a_n, q_n\} = \frac{\mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\} \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n \cap \omega = G|\mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\} \mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\}} \\ &= \frac{\mathbb{P}\{a_n \cap \omega = G|b_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\}} \end{aligned}$$

where the final two steps follow from Equation (15) and the definition of b_n .

This completes the proof. $\square$

Note that we have shown that the public belief is Markovian adapting the arguments in Banerjee (1992) and Bikhchandani et al. (1992). The modifications were required because of the differences in selections of the agents' precisions caused by the introduction of the central planner.

C PROOFS

C.1 DERIVATION OF EQUATION (4)

Lemma 7.

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \min(b_i, 1 - b_i, 1 - q_i)$$

Proof. We prove this for each case of a_i delineated in Equation (2). In the second case, we can reason as follows:

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq G|b_i, q_i) = 1 - b_i$$

Similarly, in the third case of Equation (2), we obtain:

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq B|b_i, q_i) = b_i$$

For the case when $1 - q_i \leq b_i \leq q_i$ and $a_i = s_i$ from Equation (2), we can reason as follows:

$$\begin{aligned} \mathbb{P}(a_i \neq \omega | b_i, q_i) &= \sum_{a \in \{G, B\}} \mathbb{P}(a_i = a | b_i, q_i) \mathbb{P}(\omega \neq a | b_i, q_i, a_i = a) \\ &= \mathbb{P}(s_i = G | b_i, q_i) \mathbb{P}(\omega \neq G | b_i, q_i, s_i = G) \\ &\quad + \mathbb{P}(s_i = B | b_i, q_i) \mathbb{P}(\omega \neq B | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqn 2} \quad (18)$$

$$\begin{aligned} &= y(b_i, q_i) \mathbb{P}(\omega = B | b_i, q_i, s_i = G) \\ &\quad + z(b_i, q_i) \mathbb{P}(\omega = G | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqns 11, 12, Thm 6} \quad (19)$$

$$= \begin{cases} y(0, q_i) & b_i = 0 \\ y(b_i, q_i)(1 - \frac{b_i q_i}{y(b_i, q_i)}) + z(b_i, q_i) \frac{b_i(1-q_i)}{z(b_i, q_i)} & b_i \neq 0 \end{cases} \quad \text{Eqn 13} \quad (20)$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i(1 - q_i) & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i - b_i q_i & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ 1 - q_i & b_i \neq 0 \end{cases}$$

Combining the three cases yields the claim. $\square$

C.2 PROOF OF THEOREM 1: MYOPIC ALTRUISTIC POLICY

Applying Equation (4) to $r_A(b, q)$ in Equation (5) allows us to reason as follows

$$\frac{\partial r_A(b, q)}{\partial q} = \begin{cases} -\dot{\beta}(q) & q < \max(b, 1 - b) \\ -\dot{\beta}(q) + C & q \geq \max(b, 1 - b) \end{cases} \quad (21)$$

As stated in Section 3.2, we can restrict the altruistic planner to precision $q \in [p, 1]$ because social welfare is increasing in precision (see Appendix C.9). Accordingly, $\beta(p) = 0$.

Because $\beta(\cdot)$ is increasing, the first case of Equation (21) leads to optimal precision p if $p < \max(b, 1 - b)$. That is, if the signal is not precise enough to influence the agent's action, then the planner will not expend resources for it. In the latter case, because $\beta(\cdot)$ is increasing and concave, $\dot{\beta}(\cdot)$ is positive and decreasing. Thus, $r_A(b, q)$, if the optimal lies in this range it must be attained at $q = 1$.

We now identify the regime in which the latter case is optimal.

$$\begin{aligned} r_A(b, p) &< r_A(b, 1) \\ -C \min(b, 1 - b, 1 - p) &< -\beta(1) \\ \min(b, 1 - b, 1 - p) &> \frac{\beta(1)}{C} \end{aligned}$$

If $1 - p \leq \frac{\beta(1)}{C}$, the above inequality is never satisfied, and it is never optimal to select precision 1. Otherwise, the inequality above yields $b \in \left[\frac{\beta(1)}{C}, 1 - \frac{\beta(1)}{C} \right]$. This yields Theorem 1. $\blacksquare$

C.3 PROOF OF THEOREM 2: ALTRUISTIC VALUE FUNCTION CONVEXITY

Proof Sketch We inductively prove that the expected k -th stage reward, i.e., the expected utility of the planner from the control and action of the k -th agent, is convex. The instantaneous reward (Equation (5)), which is convex with respect to public belief, provides our base case.

The first challenge encountered is the unusual nature of the public belief update. Although the state space is uncountably infinite, the belief update only takes support on a maximum of 2 values. To manage this, we define a decision tree, i.e., the complete binary tree of all possible trajectories once an initial belief and policy are fixed. Each node has two children corresponding to each possible signal realization the next agent might receive. The root is the expected instantaneous reward at time 1, i.e., from the first agent's action. The induction moves down the levels of this tree with the k -th level containing 2^{k-1} nodes, each associated with a sequence of realizations of $k - 1$ signals.

We then show that for a node in the $(k - 1)$ -th level that has convex expected reward, its two children in the k -th level satisfy the same property. This is where we must deal with the dependence of agents' actions on public belief. Note that even when applying the same precision and receiving the same signal realization, two agents beginning at different public beliefs may take opposing actions (see Equation (2)). Thus, standard results that provide easy ways of bounding the future terms of the Markov process do not apply. Here, our specific belief update is actually helpful. We can leverage the fact that Bayesian updates are martingales (i.e., $\mathbb{E}[b_{i+1}] = b_i$). Along with the convexity of instantaneous rewards, this allows us to complete the inductive step and, subsequently, the proof.

Preliminaries We first prove the following lemma containing a few useful properties of the optimal value function $V_A^*(\cdot)$.

Lemma 8.

$V_A^*(\cdot)$ is non-positive, $V_A^*(b) = V_A^*(1 - b) \forall b \in [0, 1]$, and $V_A^*(0) = V_A^*(1) = 0$ with $\pi_A^*(0) = \pi_A^*(1) = p$.

Proof.

Non-positivity

$$r(b_i, q_i) \leq 0 \quad \forall b_i, q_i \rightarrow V_A^* \leq 0$$

Symmetry

Let $n_i = 1 - b_i$ and let Λ_n and Λ_b be the decision trees rooted at states n_0 and b_0 , respectively. Showing that Λ_n and Λ_b are symmetric suffices to show the symmetry of $V^*(\cdot)$. This will be accomplished by showing that the expected rewards, belief state values, and branching probabilities of the trees are reflections of one another.

First consider the instantaneous reward function r_A . Note that the decision function of the agents when choosing their actions is symmetric from Equation (2), thus, $\mathbb{P}\{a_i = G|b_i\} = \mathbb{P}\{a_i = B|n_i\}$. Applying this, we obtain

$$\begin{aligned} r(n_i, q_i) &= -\beta(q_i) - Cn_i \mathbb{P}\{a_i = B|\omega = G, n_i\} - C(1 - n_i) \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = B|\omega = G, n_i\} - Cb_i \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = G|\omega = B, b_i\} - Cb_i \mathbb{P}\{a_i = B|\omega = G, b_i\} \\ &= r(b_i, q_i) \end{aligned}$$

Thus, the instantaneous reward function is symmetric about 0.5.

Now consider the belief state update when beginning with belief n_i assuming signal precision q_i . we will use $\bar{x}$ and $\underline{x}$ to represent updated beliefs from x after observing action G or B , respectively. From Equation (13):

$$\begin{aligned} \bar{n}_i &= \frac{q_i}{y(n_i, q_i)} n_i = \frac{q_i}{z(b_i, q_i)} (1 - b_i) = 1 - \frac{1 - q_i}{z(b_i, q_i)} b_i = 1 - \bar{b}_i \\ \underline{n}_i &= \frac{1 - q_i}{z(n_i, q_i)} n_i = \frac{q_i}{y(b_i, q_i)} (1 - b_i) = 1 - \frac{q_i}{y(b_i, q_i)} b_i = 1 - \bar{b}_i \end{aligned}$$

Thus, the belief updates are also symmetric. Combining this with the symmetry of the reward we obtain:

$$\begin{aligned} r(\bar{n}_i) &= r(1 - \bar{b}_i) = r(\bar{b}_i) \\ r(\underline{n}_i) &= r(1 - \bar{b}_i) = r(\bar{b}_i) \end{aligned}$$

Thus, the expected rewards and belief state values of Λ_n and Λ_b are mirror images of one another.

Finally, consider the branching probabilities which we will denote as $y(b, q) = 1 + 2bq - b - q$ for observing action G and $z(b, q) = 1 - y(b, q)$ for observing action B .

$$\begin{aligned} y(n, q) &= 1 + 2nq - n - q = 1 + 2(1 - b)q - (1 - b) - q = b + q - 2bq = z(b, q) \\ z(n, q) &= 1 - y(n, q) = 1 - z(b, q) = y(b, q) \end{aligned}$$

Thus, the branching probabilities of Λ_n and Λ_b are also symmetric, completing the proof.

Extremal values

We will show that $V_A^*(0) = 0$ and rely on the symmetry about 0.5 proven above to show that $V_A^*(1) = 0$.

From Equation (3), when $b_0 = 0$, $b_0^+ = b_0^- = 0$ for any value of precision. Thus, regardless of signal realizations or control actions, the agents' actions and, subsequently, the public belief, will remain unchanged. That is, $b_i = 0 \forall i$ and, from Equation (2), $a_i = B \forall i$.

From Equation (5), we can then write:

$$\begin{aligned}
V_A^*(0) &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(b_i, \pi(b_i)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(0, \pi(0)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(\pi(0)) - C b_i \mathbb{P}\{a_i = B | \omega = G, b_i = 0\} - C(1 - b_i) \mathbb{P}\{a_i = G | \omega = B, b_i = 0\}] \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(q_i)] \right]
\end{aligned} \tag{22}$$

Thus, to attain $V_A^*(0)$, it suffices to choose $\pi(0)$ such that $\beta(\pi(0))$ is minimized for each i . Recall that $\beta(\cdot)$ is increasing on $[0.5, 1]$ and takes its minimum at $\beta(p) = 0$. Thus, by choosing $q_i = p \forall i$, Equation 22 yields $V_A^*(0) = 0$ with corresponding optimal policy $\pi_A^*(0) = p$.

The aforementioned symmetry about 0.5, shows that $V_A^*(1) = 0$, completing the proof. $\square$

We now introduce the concept of *decision trees* for Markovian and more general policies. A policy can be represented by a decision tree, which is a complete binary tree as in figure 3. Every node of the decision tree represents the state of the system, i.e., the corresponding belief value when the controller sends its private signal to the agent corresponding to the level of the node. The branches represent the actions taken by the agent corresponding to the level once it receives its private signal. The controller's policies (i.e. choice of precision) determine the probabilities of the actions from each node and the belief resulting from the action. The probabilities of an action at a certain node can be considered the weight of the branch. A decision tree is uniquely identified by the belief values associated with nodes and the weights of the branches. For a deterministic stationary Markovian policy, the controller's choice of precision, and therefore the probability associated with an action, is a deterministic function of the belief value of the node, regardless of the level of the node in the tree and the path to the node from the root. Thus, the decision tree of any such policy, i.e. the belief values at the nodes and the weights of the branches, is determined entirely by the value of the initial state. For an arbitrary policy of the controller, the controller's decision at a given node can depend on the entire path leading to the node.

Proof of Theorem 2 We now prove the convexity of $V_A^*(\cdot)$ with respect to public belief.

Proof.

We will first show that, for fixed precision, the instantaneous reward r_A is convex in public belief. We can rewrite r_A as follows using Lemma 7:

$$r_A(x, q) = -\beta(q) - C \min(x, 1 - x, 1 - q) \tag{23}$$

The pointwise minimum of linear functions is concave. As we are subtracting the minimum, the overall function is convex with respect to x .

We consider arbitrary x_0, m_0, n_0 such that $tm_0 + (1 - t)n_0 = x_0$ for some $t \in [0, 1]$. To show the convexity of $V_A^*(\cdot)$ we must show that $V_A^*(x_0) \leq tV_A^*(m_0) + (1 - t)V_A^*(n_0)$. By definition of the optimal value function, $V_A^*(x) \geq V_A^\pi(x)$ for any policy π . Thus, it will suffice to show that there exist $\tilde{\pi}$ such that

$$V_A^*(x_0) \leq tV_A^{\tilde{\pi}}(m_0) + (1 - t)V_A^{\tilde{\pi}}(n_0) \tag{24}$$

We refer to the decision tree of π^* when the initial state is x_0 as Λ .

We consider $\tilde{\pi}$ to be a policy of the controller that chooses the precision at each node to be the same as what Λ does at the counterpart node. The nodes of decision trees are considered to be counterparts if they can be reached via the same sequence of actions from the root (see figure 3 for illustration). Thus, the precision chosen by this policy is the same as what Λ does after the same set of actions of the agents. We refer to the decision trees for $\tilde{\pi}$ with initial state m_0 (respectively, n_0) as Λ_m (Λ_n).

We now illustrate the policy $\tilde{\pi}$ for initial values m_0 and n_0 considering the decision trees Λ_m and Λ_n . While doing so, we illustrate counterpart nodes and the weights of the branches. We will select one path of actions λ out of these decision trees by specifying set $\mathcal{G}$ as the epochs where action G is taken and $\mathcal{B}$ as its complement. Refer to figure 3 for illustration. Considering the decision tree Λ , we will refer to x_i as the state at epoch i when beginning with belief x_0 and following π^* along this path of actions. That is, x_i assumes one of 2^i possible belief values at epoch i starting from x_0 . In decision tree Λ_m (respectively, Λ_n), we use m_i (n_i) to denote the

public belief after beginning with m_0 (n_0), following the path λ . Node m_i (respectively n_i) in decision tree Λ_m (Λ_n) is the counterpart of node x_i from decision tree Λ . Note that, $\tilde{\pi}(m_i)$ can now be illustrated notationally. Specifically, $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$. Similarly, $\tilde{\pi}$ can be defined at each node of Λ_m , considering the action taken by π^* at each counterpart node in Λ . Note that $y(x_i, \pi_A^*(x_i))$ (respectively, $z(x_i, \pi_A^*(x_i))$) is the weight of the branch corresponding to G (B) emanating from x_i in Λ . Because $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$, $y(m_i, \tilde{\pi}(m_i)) = y(m_i, \pi_A^*(x_i))$ is the weight of the branch emanating from m_i for action G in Λ_m . This extends similarly to branches associated with action B and Λ_n . Thus, the three decision trees differ in both the values of the nodes and the weights of the branches. Since π^* is deterministic, so is $\tilde{\pi}$. Therefore, from Equation (13) and Equation (3), x_i , m_i , and n_i are deterministic quantities once the specific path of actions is specified by $\mathcal{G}$.

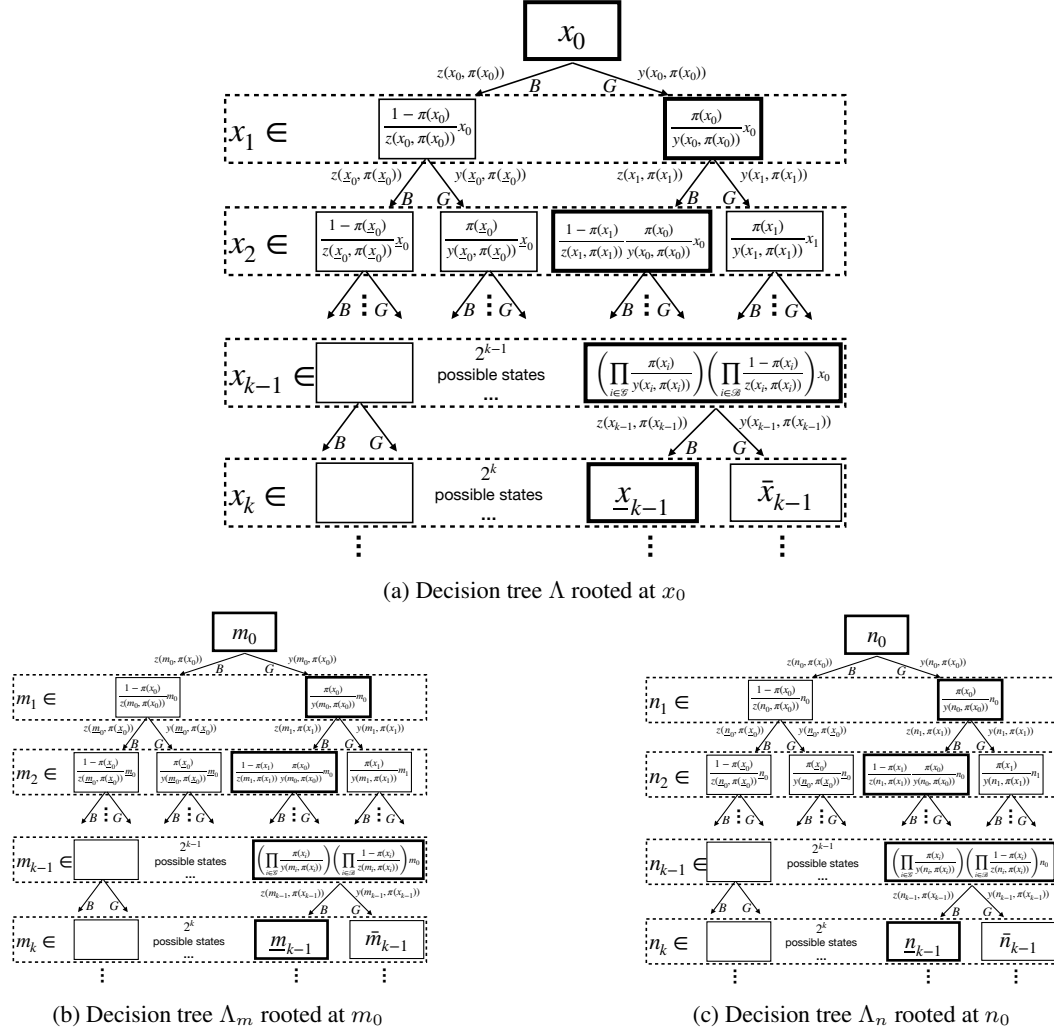


Figure 3: Here we depict the belief evolution in the form of three binary trees. The tree begins with state x_0 , m_0 , and n_0 , respectively, and level i contains 2^i possible states reachable after i epochs. Left branches correspond to action B leading to updated belief $\underline{x}_0$ (respectively, $\underline{m}_0$ and $\underline{n}_0$). Right branches correspond to action G leading to updated belief $\bar{x}_0$ (respectively, $\bar{m}_0$ and $\bar{n}_0$). Each branch is labeled with the corresponding probability, and the bold states indicate an example path, like λ . In the example path, $0 \in \mathcal{G}$ and $1, k-1 \in \mathcal{B}$. The bold states in each level of the three trees are counterparts.

Now let $\phi_k^\pi(b_0) = \mathbb{E}[r_A(b_k, \pi(b_k))]$ be the expected instantaneous reward at epoch k when beginning with state b_0 and applying policy π and $\phi^*(x) = \phi^{\pi^*}(x)$. Applying Fubini's Theorem, we can rewrite our value function

as

$$V_A^\pi(b_0) = \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r_A(b_i, \pi(b_i)) \right] = \sum_{i=0}^{\infty} \delta^i \mathbb{E}[r_A(b_i, \pi(b_i))] = \sum_{i=0}^{\infty} \delta^i \phi_i^\pi(b_0) \quad (25)$$

Thus, we can rewrite (24), as

$$\sum_{i=0}^{\infty} \delta^i \phi_i^*(x_0) \leq t \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(m_0) \right] + (1-t) \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(n_0) \right] \quad (26)$$

To prove this, we show that for all $i = 0, 1, 2, \dots$

$$\phi_i^*(x_0) \leq t \phi_i^{\tilde{\pi}}(m_0) + (1-t) \phi_i^{\tilde{\pi}}(n_0) \quad (27)$$

leading directly to (26), (24), and finally the convexity of $V^*(\cdot)$ with respect to public belief.

Now let us consider general $i = k$. $\phi_{k-1}^\pi(x_0)$ will have 2^{k-1} terms each corresponding to one path of possible actions in the set $\{G, B\}^{k-2}$. Thus, at epoch $k-1$, the path λ will yield the following term in $\phi_{k-1}^*(x_0)$

$$\left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right) r_A(x_{k-1}, \pi_A^*(x_{k-1})) \quad (28)$$

We now define $P_\lambda^*(x_0) = \left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right)$, that is, the probability of following the specified path λ when beginning with state x_0 and following policy π_A^* . Again, we will use $\bar{x}_i$ and $\underline{x}_i$ to represent updated beliefs from x_i after observing action G or B , respectively. Thus, $\bar{x}_i$ and $\underline{x}_i$ constitute belief values of the children nodes (in the $(i+1)$ th level) of the node with belief value x_i in the i -th level. Refer to levels $k-1$ and k in figure 3a. Considering the children $\bar{x}_{k-1}$ and $\underline{x}_{k-1}$ (in the k -th level) of the node x_{k-1} (in the $(k-1)$ -th level), we obtain the following two terms in $\phi_k^*(x_0)$:

$$P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (29)$$

Similarly, we define $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \tilde{\pi}(m_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \tilde{\pi}(m_i)) \right)$, that is, the probability of following the specified path λ when beginning with state m_0 and following policy $\tilde{\pi}$. Because $\tilde{\pi}(m_i) = \pi_A^*(x_i)$, $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \pi_A^*(x_i)) \right)$. We also define $P_\lambda^{\tilde{\pi}}(n_0)$ by replacing $\{m_i\}$ with $\{n_i\}$.

The corresponding terms in $\phi_k^{\tilde{\pi}}(m_0)$ and $\phi_k^{\tilde{\pi}}(n_0)$ are

$$P_\lambda^*(m_0) \left[y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (30)$$

$$P_\lambda^{\tilde{\pi}}(n_0) \left[y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (31)$$

We will show that

$$\begin{aligned} & P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & \leq t P_\lambda^*(m_0) \left[y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & + (1-t) P_\lambda^{\tilde{\pi}}(n_0) \left[y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \end{aligned} \quad (32)$$

Similarly, each pair of children for each term in the $(k-1)$ -th level of the belief tree can be combined and the corresponding inequalities can be obtained following the same process. Summing the resulting inequalities will yield (27), leading to (26), (24), and the convexity of $V^*(\cdot)$.

Proof of (32)

We now focus on the latter of the two terms of (29). We will show the following two properties

$$\begin{aligned} \underline{x}_{k-1} &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \end{aligned} \quad (33)$$

$$\begin{aligned}
1 &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
&+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)
\end{aligned} \tag{34}$$

Defining $\tilde{t} = t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)$, it follows from (33) and (34) that

$$\underline{x}_{k-1} = \tilde{t} \underline{m}_{k-1} + (1-\tilde{t}) \underline{n}_{k-1} \tag{35}$$

From the expressions for $\tilde{t}$, $P_\lambda^*(x_0)$, $P_\lambda^{\tilde{\pi}}(m_0)$, it follows that:

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = t z(m_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(m_0) \tag{36}$$

Similarly, we can show

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = (1-t) z(n_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(n_0) \tag{37}$$

We apply this as follows to the latter term of (29):

$$\begin{aligned}
&P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \\
&\leq P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) \\
&\quad [\tilde{t} r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) + (1-\tilde{t}) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Convexity of } r, \text{Eqn 35}) \\
&= t [P_\lambda^{\tilde{\pi}}(m_0) z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\
&\quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Eqns 36, 37})
\end{aligned} \tag{38}$$

Applying the same argument to the first term of (29) yields:

$$\begin{aligned}
&P_\lambda^*(x_0) y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \\
&\leq t [P_\lambda^{\tilde{\pi}}(m_0) y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1}))] \\
&\quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1}))]
\end{aligned} \tag{39}$$

Summing (38) and (39) yields (32).

We now complete the proof of (32) and, therefore, the entire proof, by proving (33) and (34).

Proof of (33)

From the recurrence relation in Equation (13) and Equation (3), we can write $\underline{x}_{k-1}$, $\underline{m}_{k-1}$, and $\underline{n}_{k-1}$ as the product of all belief updates with x_0 , m_0 , and n_0 , respectively, as follows. Each term in the products corresponds to the multiplicative state update after a single action.

$$\underline{x}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) x_0 \tag{40}$$

$$\underline{m}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(m_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(m_i, \pi_A^*(x_i))} \right) m_0 \tag{41}$$

$$\underline{n}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(n_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(n_i, \pi_A^*(x_i))} \right) n_0 \tag{42}$$

These equations may be best understood via the illustration in figure 3. We then reorganize terms to relate $\underline{x}_{k-1}$ to $\underline{m}_{k-1}$ and $\underline{n}_{k-1}$, using the fact that $x_0 = tm_0 + (1-t)n_0$.

$$\begin{aligned}\underline{x}_{k-1} &= \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) [tm_0 + (1-t)n_0] \quad (\text{from 40}) \\ &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &\quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \quad (\text{from 41 and 42}) \quad (43)\end{aligned}$$

This proves (33).

Proof of (34)

We first consider the numerator of the first term in (34): $\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(m_i, \pi_A^*(x_i))$. We now prove that this can be written as $\alpha_0 m_0 + \beta_0$ for some α_0, β_0 which depend only on $\{\pi_A^*(x_i)\}_{i=1, \dots, k-1}$ and the nature of the dependence is determined by the sequence of actions along this path in stages 1 to $k-1$. We will prove this via induction working backwards from the final term (i.e. the $(k-1)$ -th term) of the product. Therefore our base case will be the final term of this product which can be either $y(m_{k-1}, \pi_A^*(x_{k-1}))$ or $z(m_{k-1}, \pi_A^*(x_{k-1}))$, depending on the preceding action (in the $(k-1)$ -th stage).

$$\begin{aligned}y(m_{k-1}, \pi_A^*(x_{k-1})) &= 1 + 2m_{k-1}\pi_A^*(x_{k-1}) - m_{k-1} - \pi_A^*(x_{k-1}) \\ &= (2\pi_A^*(x_{k-1}) - 1)m_{k-1} + (1 - \pi_A^*(x_{k-1})) \\ z(m_{k-1}, \pi_A^*(x_{k-1})) &= m_{k-1} + \pi_A^*(x_{k-1}) - 2m_{k-1}\pi_A^*(x_{k-1}) \\ &= (1 - 2\pi_A^*(x_{k-1}))m_{k-1} + (\pi_A^*(x_{k-1}))\end{aligned}$$

Thus, the final term of the product is of the form $\alpha_{k-1}m_{k-1} + \beta_{k-1}$ where α_{k-1} and β_{k-1} depend only upon $\pi_A^*(x_{k-1})$, and the nature of the dependence is determined entirely by the $(k-1)$ -th action (since this action determines whether the final term is $y(\cdot)$ or $z(\cdot)$).

For the inductive step, assume that $\prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_{j+1}m_{j+1} + \beta_{j+1}$ for some $\alpha_{j+1}, \beta_{j+1}$ which depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the nature of the dependence is determined entirely by the sequence of actions in this path for $i > j$ to $i = k-1$. We will now show that $\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_j m_j + \beta_j$ again with α_j, β_j depending only on $\{\pi_A^*(x_i)\}_{i > j-1}$ with the nature of the dependence determined by the sequence of actions on

this path for $i > j-1$ to $i = k-1$. We define $f(b, q, a) = \begin{cases} y(b, q) & a = G \\ z(b, q) & a = B \end{cases}$ and $g(q, a) = \begin{cases} q & a = G \\ 1 - q & a = B \end{cases}$.

Thus,

$$\begin{aligned}&\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) [\alpha_{j+1}m_{j+1} + \beta_{j+1}] \quad (\text{inductive hypothesis}) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \left[\alpha_{j+1} \frac{g(\pi_A^*(x_j), a_{j+1})}{f(m_j, \pi_A^*(x_j), a_{j+1})} m_j + \beta_{j+1} \right] \quad (\text{Equation (13)}) \\ &= \alpha_{j+1}g(\pi_A^*(x_j), a_{j+1})m_j + \beta_{j+1}f(m_j, \pi_A^*(x_j), a_{j+1})\end{aligned}$$

Thus, from the definitions of $f(\cdot), g(\cdot), y(\cdot)$, and $z(\cdot)$,

$$\alpha_j = \begin{cases} \alpha_{j+1}\pi_A^*(x_j) + \beta_{j+1}(2\pi_A^*(x_j) - 1) & a_{j+1} = G \\ \alpha_{j+1}(1 - \pi_A^*(x_j)) + \beta_{j+1}(1 - 2\pi_A^*(x_j)) & a_{j+1} = B \end{cases} \text{ and } \beta_j = \begin{cases} \beta_{j+1}(1 - \pi_A^*(x_j)) & a_{j+1} = G \\ \beta_{j+1}\pi_A^*(x_j) & a_{j+1} = B \end{cases}$$

In both cases, by our inductive hypothesis, $\alpha_{j+1}, \beta_{j+1}$ depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the actions in this path from $i > j$ to $i = k-1$. Thus, the claim holds for α_j, β_j , completing the induction.

Now recall that the denominator of (34) is $\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(x_i, \pi_A^*(x_i))$. Note that the policy and sequence of actions is the same as those for the numerator. Thus, because, as shown, the coefficients α_0, β_0 depend only on $\{\pi_A^*(x_i)\}$ and the path of actions, the same logic can be applied to write the denominator

as $\alpha_0 x_0 + \beta_0$ for the same α_0, β_0 . This can be done similarly for the numerator of the second term in (34). Thus, we can reason about the coefficients in (33) as follows:

$$\begin{aligned}
& t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& = \frac{t(\alpha_0 m_0 + \beta_0) + (1-t)(\alpha_0 n_0 + \beta_0)}{\alpha_0 x_0 + \beta_0} \\
& = \frac{\alpha_0 x_0 + \beta_0}{\alpha_0 x_0 + \beta_0} \quad (\text{Since } x_0 = t m_0 + (1-t) n_0) \\
& = 1
\end{aligned}$$

Thus, (34) holds, completing the proof. $\square$

C.4 ALTRUISTIC MYOPIC POLICY BOUND

Lemma 9.

$$\pi_A^0(b) \leq \pi_A^*(b) \quad \forall b \in [0, 1]$$

Assume by way of contradiction that there exists $b \in [0, 1]$ such that $\pi_A^0(b) > \pi_A^*(b)$. We will show that $V_A^*(b) < V_A^\pi(b)$ for a policy π we construct, violating the optimality of $V_A^*(\cdot)$.

Let π be such that, when starting at public belief b , the policy π applies precision $\pi_A^0(b)$ at the current time step and then applies the optimal policy at all future time steps. We now elaborate $V_A^*(b) - V_A^\pi(b)$ by breaking the value function into instantaneous reward r_A and future discounted reward. We will refer to b^+ and b^- as the updated beliefs after receiving signals G and B , respectively when applying the optimal policy and b_m^+ and b_m^- as the same when applying the myopic policy.

$$\begin{aligned}
V_A^*(b) - V_A^\pi(b) &= r_A(b, \pi_A^*(b)) - r_A(b, \pi_A^0(b)) \\
&+ \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \\
&< \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \quad (\text{definition of } \pi_A^0) \\
&\leq 0 \quad (\text{Theorem 2})
\end{aligned}$$

Here, we applied the fact that the instantaneous reward under $\pi_A^0(\cdot)$ must be greater from the definition of the myopic optimal policy.

We then relied on the convexity of $V_A^*(\cdot)$ from Theorem 2 for the future cost. From Equation (13), note that $b_m^- \leq b^- \leq b \leq b^+ \leq b_m^+$. Along with this convexity, this leads to the last step above and completes the proof. $\blacksquare$

C.5 PROOF OF THEOREM 3: OPTIMAL ALTRUISTIC POLICY

Consider $V_A^\pi(b)$ with policy π applying precision q in the current time step and applying the optimal policy at all future time steps. The q that maximizes $V_A^\pi(b)$ provides the optimum precision starting at public belief b . Because the planner never benefits from decreasing precision below the baseline p (see Appendix C.9), we restrict q to $[p, 1]$ without loss of generality.

We will let b^+ and b^- denote the positive and negative belief updates possible after receiving a signal with precision q . Taking the second derivative of $V_A^\pi(b)$ with respect to q for $q \geq \max(b, 1-b)$:

$$\frac{\partial^2}{\partial q^2} V_A^\pi(b) = \begin{cases} -\ddot{\beta}(q) + \delta b^2 (1-b)^2 \left[\frac{\ddot{V}_A^*(b^+)}{y^3(b, q)} + \frac{\ddot{V}_A^*(b^-)}{z^3(b, q)} \right] & q \geq \max(b, 1-b) \\ -\ddot{\beta}(q) & \text{otherwise} \end{cases}$$

With the convexity and concavity of $V_A^*(\cdot)$ and $\beta(\cdot)$, respectively, the above shows that $V_A^\pi(b)$ is convex with respect to q on $[\max(b, 1-b), 1]$. Thus, on this interval, it is maximized at one of the two extreme points.

For $q \in [0.5, \max(b, 1 - b))$, the derivative of $V_A^\pi(b)$ with respect to q is negative; thus, the optimal choice will be the baseline precision p to minimize the cost. Thus, there are three possible optimal precisions: p , $\max(b, 1 - b)$, and 1.

Note that the expected value under the latter two candidates is always strictly negative. Under precision p , however, the expected value is 0 at $b = 0$ and $b = 1$. This, combined with the non-positivity and symmetry of the value function (Lemma 8) implies the existence of d_A such that $\pi_A^*(b) = p$ for $b \in [0, d_A] \cup (1 - d_A, 1]$. Furthermore, from Theorem 1 and applying Lemma 9, $d_A \in (0, t_M)$ where t_M is defined as in the statement of Theorem 1.

Finally, Theorem 1 and applying Lemma 9, also imply the existence of $t_A \in [d_A, t_M]$ such that $\pi_A^*(b) = 1$ for $b \in (t_A, 1 - t_A)$. This completes the proof. $\blacksquare$

C.6 PROOF OF THEOREM 4: MYOPIC BIASED POLICY

Preliminaries Applying the action rule from (2), the instantaneous reward (6) can be written as follows:

$$r_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) - C & b < 1 - q \\ -\beta(|q - p|) & b > q \end{cases} \quad (44)$$

Note that $z(b, q) = b + q - 2bq$ increases with respect to q when $b < 0.5$ and decreases with respect to q when $b > 0.5$.

At $b = 0.5$, $z(b, q) = z(0.5, q) = 0.5$ regardless of the precision chosen. Furthermore, at this public belief, $q \geq \max(b, 1 - b)$ because $q \geq 0.5$ for any q , falling in the first case of (44). In this case, the second term of $r_B(b, q)$ is unchanging with respect to q , so the planner seeks to maximize only the first term, which is accomplished at $q = p$.

We now prove a series of lemmas (10-14) which will assist us in our proof of Theorem 4.

Lemma 10.

If $\pi_B^0(b) \geq \max(b, 1 - b)$, then $\pi_B^0(b) \in \{\max(b, 1 - b), p, 1\}$.

Proof. The reward is differentiable everywhere except $q = p$. Thus, we can write its derivative with respect to the chosen precision as follows (except at $q = p$):

$$\frac{\partial r_B(b, q)}{\partial q} = \begin{cases} -\text{sign}(q - p)\dot{\beta}(|q - p|) & q \geq \max(b, 1 - b) \\ -C(1 - 2b) & \\ -\text{sign}(q - p)\dot{\beta}(|q - p|) & \text{otherwise} \end{cases} \quad (45)$$

In the former case, we can write the second derivative of the reward with respect to q as follows, for $q \neq p$:

$$\frac{\partial^2}{\partial q^2} r_B(b, q) = -\ddot{\beta}(|q - p|)$$

Because $\beta(\cdot)$ is concave, this expression is non-negative, and $r_B(b, q)$ is convex with respect to q . Thus, the reward is maximized at one of the extreme points $\{\max(b, 1 - b), 1\}$ or p because $r_B(b, q)$ is not differentiable at $q = p$. $\square$

Lemma 11.

For $b \in [0, 1 - p)$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. Because $p > 0.5$, for $b \in [0, 1 - p)$, $\max(b, 1 - b) = 1 - b > p$.

First, consider the case when $\pi_B^0(b) < \max(b, 1 - b) = 1 - b$. From (44), in such instances, the reward is maximized by minimizing $|q - p|$. Thus, the optimal precision lower than $\max(b, 1 - b)$ is p .

Now consider $\pi_B^0(b) \geq \max(b, 1 - b) = 1 - b$. By Lemma 10, $\pi_B^0(b) \in \{1 - b, p, 1\}$. What remains is to rule out $\pi_B^0(b) = 1$. To do so, we compare $r_B(b, 1)$ and $r_B(b, 1 - b)$. Both fall under the first case of (44). Because $\beta(\cdot)$ is increasing in its argument and $p < 1 - b \leq 1$, the first term of the reward from (44) is greater for precision $1 - b$ than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, $r_B(b, 1) \leq r_B(b, \max(b, 1 - b))$ so we can rule out precision 1. This completes the proof. $\square$

Lemma 12.

For $b \in (p, 1]$, $\pi_B^0(b) = p$.

Proof. The reward is strictly negative for any $q \neq p$ because of the term $-\beta(|q - p|)$. For $b \in (p, 1]$ and $q = p$, $p < \max(b, 1 - b) = b$, leading to the third case of (44) and yielding reward 0. Thus, the optimal precision is p . $\square$

Lemma 13.

For $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. When $p = 0.5$, the interval $[1 - p, 0.5]$ contains only $\{0.5\}$. At $b = 0.5$ $p = 1 - b$, thus $\pi_B^0(b) = p = 1 - b = 0.5$. We deal with $p > 0.5 \geq b$ for the remainder of the proof.

For $b \in [1 - p, 0.5]$, $\max(b, 1 - b) = 1 - b \leq p$. Thus, from the first case of (44), precision p yields reward $r_B(b, p) = -Cz(b, p)$. We now consider two cases: $q < \max(b, 1 - b)$ and $q \geq \max(b, 1 - b)$.

Suppose $q < \max(b, 1 - b) = 1 - b$. This leads to the second case of (44) and yields reward $-\beta(|q - p|) - C$. Furthermore, $z(b, q)$ is increasing with respect to q for $b < 0.5$ and takes values between $z(b, 0.5) = 0.5$ and $z(b, 1) = 1 - b$. Thus, $z(b, q) < 1$ for all q . Therefore, the reward from policy $q < 1 - b$ is strictly lower than $r_B(b, p)$, and we can rule out this case.

When $q \geq \max(b, 1 - b)$, applying Lemma 10 yields that the remaining candidate precisions are $\{1 - b, p, 1\}$. We rule out precision 1 by comparing it to precision p . The first term of the reward from (44) is greater for precision p than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, the second term is also greater for precision p , and we can rule out precision 1. This completes the proof. $\square$

We now define the notion of ϵ -optimal policies.

Definition 1. ϵ -optimal policies

An ϵ -optimal policy $\pi^\epsilon(b)$ is any policy such that the following holds

$$V^*(b) \leq V^{\pi^\epsilon}(b) + \epsilon$$

for $\epsilon > 0$.

Note that from the definition, any optimal policy is also ϵ -optimal for any $\epsilon > 0$. Thus, $\pi_B^0(b)$ is also ϵ -optimal

Following this definition, let $\pi_B^{\epsilon, 0}(b)$ denote an ϵ -optimal myopic policy for the biased planner.

Lemma 14.

For $b \in (0.5, p]$ and any $\epsilon > 0$, there are two possibilities:

1. $\pi_B^0(b) \in \{p, 1\}$, or

2. $\pi_B^{\epsilon, 0}(b) = b - \epsilon$

Proof. For $b \in (0.5, p]$, $\max(b, 1 - b) = b \leq p$. We consider two cases: $q \geq \max(b, 1 - b)$ or $q < \max(b, 1 - b)$.

In the first case, Lemma 10 yields that $\pi_B^0(b) \in \{b, p, 1\}$. We rule out precision b by comparing it to precision p . The first terms of the reward from (44) is greater for precision p . Furthermore, since $z(b, q)$ is decreasing with respect to q for $b > 0.5$, the second term of the reward is also greater for precision p . Thus, we can rule out precision b . This yields the first claim.

In the latter case, $q < \max(b, 1 - b) = b$. Therefore, (1) since $q \geq 0.5$, $q \in [0.5, b)$, and (2) the third case of (44) applies leading to reward $-\beta(|q - p|)$. Thus, the reward $r_B(b, q)$ is an increasing function of $q \in [0.5, b)$ because $0.5 < b \leq p$. This leads directly to the second claim. $\square$

Remark 3. As noted in Puterman (1990) Section 2.3.1, an optimal policy may not exist for infinite action spaces because the supremum may not be attained. When an optimal policy does not exist, we instead seek ϵ -optimal policies, i.e., policies that yield reward within $\epsilon > 0$ of the supremum. In particular, this is true for case 2 of Lemma 14.

Proof of Theorem 4 We consider the myopic biased policy in 4 cases: $b \in [0, 1 - p]$, $b \in [1 - p, 0.5]$, $b \in (0.5, p]$, and $b \in (p, 1]$, making use of the fact that $p \geq 0.5$. Based on Lemmas 11-14, these cases have the following candidate policies: $\{p, 1 - b\}$, $\{1 - b, p\}$, $\{1, p, b - \epsilon\}$, and $\{p\}$, respectively.

Case 1: $b \in [0, 1 - p]$, $\pi_B^0(b) \in \{p, 1 - b\}$ (Lemma 11)

We will show the existence of a threshold t_1 such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

We can express the reward under the two candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, p) &= -C \\ r_B(b, 1 - b) &= -\beta(1 - b - p) - C(1 - 2b(1 - b)) \end{aligned}$$

The derivative of $r_B(b, 1 - b)$ with respect to b is

$$\frac{\partial}{\partial b} r_B(b, 1 - b) = \dot{\beta}(1 - p - b) + 2C(1 - 2b)$$

which is positive as $\beta(\cdot)$ is increasing and $b < 1 - p \leq 0.5$. Furthermore, when evaluated at $b = 0$ and $b = 1 - p$ we obtain

$$\begin{aligned} r_B(0, 1) &= -\beta(1 - p) - C < r_B(b, p) \\ r_B(1 - p, p) &= -C(1 - 2p(1 - p)) > r_B(b, p) \end{aligned}$$

Thus, $r_B(b, 1 - b)$ is increasing on $[0, 1 - p]$ with $r_B(b, 1 - b) < r_B(b, p)$ for $b = 0$ and vice versa for $b = 1 - p$. Because $\beta(\cdot)$ is continuous, so are both reward functions. Thus, there exists $t_1 \in (0, 1 - p)$ such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

Case 2: $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{1 - b, p\}$ (Lemma 13)

If $p = 0.5$, this interval contains only $\{0.5\}$, and, for $b = 0.5$, $1 - b = p$. Thus, the two candidate policies are equivalent, and the optimal policy is $\pi_B^0(b) = p$. For the remainder of this case, we consider $p > 0.5$.

We now prove the existence of t_2, t_3 such that $\pi_B^0(b) = 1 - b$ for $b \in [t_2, t_3]$, and $\pi_B^0(b) = p$ for $b \in [1 - p, t_2] \cup [t_3, 0.5]$.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1 - b) &= -\beta(p - 1 + b) - Cz(b, 1 - b) \\ r_B(b, p) &= -Cz(b, p) \end{aligned}$$

This leads to the following condition for the optimal policy to be $1 - b$:

$$\begin{aligned} r_B(b, 1 - b) &\geq r_B(b, p) \\ \beta(p - 1 + b) &\leq C[z(b, p) - z(b, 1 - b)] \\ \beta(p - 1 + b) &\leq C(p - 1 + b)(1 - 2b) \end{aligned} \tag{46}$$

The LHS of (46) is a concave, increasing function of b on $[1 - p, 0.5]$. The RHS of (46) is a concave parabola that increases in the first half of $[1 - p, 0.5]$ and decreases in the second half.

Now consider the two halves of the interval $[1 - p, 0.75 - 0.5p]$ and $[0.75 - 0.5p, 0.5]$. In the first half, the LHS and RHS of (46) are increasing and concave, with the RHS also being a parabola. Thus, in the first half, the two intersect at most two points (one of which is at $b = 1 - p$). In the second half, the LHS of (46) is increasing while the RHS is decreasing. Thus, they intersect at most once. Let the two possible intersections for $b > 1 - p$ be t_2 and t_3 with $t_2 \leq t_3$. Thus, what remains is to determine the optimal on each of $(1 - p, t_2)$, (t_2, t_3) , and $(t_3, 0.5)$. At $b = 0.5$, $r_B(b, p) > r_B(b, 1 - b)$ because $p > 0.5$, thus $\pi_B^0(b) = p$ for $b \in (t_3, 0.5)$. Subsequently, $\pi_B^0(b) = 1 - b$ for (t_2, t_3) , and $\pi_B^0(b) = p$ for $(1 - p, t_2)$.

Case 3: $b \in (0.5, p]$, $\pi_B^{\epsilon, 0}(b) \in \{1, p, b - \epsilon\}$ (Lemma 14)

In this case, we show the existence of $0.5 \leq t_4 \leq t_5 \leq p$ such that the optimal precisions on intervals $(0.5, t_4]$, (t_4, t_5) , and $[t_5, p)$ are p , 1 , and $b - \epsilon$, respectively.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1) &= -\beta(1 - p) - C(1 - b) \\ r_B(b, p) &= -Cz(b, p) \\ r_B(b, b - \epsilon) &= -\beta(p - b + \epsilon) \end{aligned}$$

When evaluated at $b = p$, $r_B(b, b - \epsilon)$ is arbitrarily close to 0, and, therefore, $b - \epsilon$ is the optimal precision on some interval $(t_5, p]$ for $0.5 \leq t_5 < p$.

Now, note that $r_B(b, 1)$ and $r_B(b, p)$ are linear functions in b while $r_B(b, b - \epsilon)$ is convex in b (from the concavity of β). Thus, $r_B(b, b - \epsilon) - r_B(b, p)$ and $r_B(b, b - \epsilon) - r_B(b, 1)$ are convex and, as a result, have convex level sets. This implies that the set on which both of these expressions are positive, i.e., the set for which $b - \epsilon$ is the optimal policy, must also be convex. Therefore, $b - \epsilon$ cannot be optimal on any subset of $(0.5, t_5)$.

What remains is to compare precisions 1 and p on the interval $(0.5, t_5]$. To do so, we inspect the values of $r_B(b, 1)$ and $r_B(b, p)$ at 0.5 and their derivatives.

$$\begin{aligned} r_B(0.5, 1) &= -\beta(1 - p) - 0.5C \\ r_B(0.5, p) &= -0.5C \end{aligned}$$

Because $r_B(0.5, 1) < r_B(0.5, p)$ there exists interval $(0.5, t_4)$ with $t_4 \leq t_5$ such that $\pi_B^0(b) = p$. Furthermore, because both are linear in b , there is at most one crossover point where their ordering changes. Thus, if $t_4 < t_5$, then $\pi_B^0(b) = 1$ on (t_4, t_5) .

Case 4: $b \in (p, 1]$, $\pi_B^0(b) = p$

This follows directly from Lemma 12 and completes the proof of Theorem 4. ■

C.7 BIASED VALUE FUNCTION MONOTONICITY

Lemma 15. For any $b_1, b_2 \in [0, 1]$ such that $b_1 \leq b_2$,

$$V_B^*(b_1) \leq V_B^*(b_2)$$

Proof. Applying (44), the derivative of the biased reward with respect to b can be written as follows:

$$\frac{\partial}{\partial b} r_B(b, q) = \begin{cases} -C(1 - 2q) & q \geq \max(b, 1 - b) \\ 0 & \text{otherwise} \end{cases} \quad (47)$$

Because $q \geq 0.5$ and $z(b, q) \in [0, 1]$, it follows that $r_B(b, q)$ is increasing with respect to b .

Next, note that the Bayesian nature of belief updates implies that the belief process is a martingale. That is, for any fixed q_i , $\mathbb{E}[b_{i+1}] = b_i$. To demonstrate this consider two cases: (1) $q_i < \max(b_i, 1 - b_i)$, and (2) $q_i \geq \max(b_i, 1 - b_i)$.

In the former case, it follows from (3) that $b_{i+1} = b_i$, satisfying the claim.

In the latter case, (13) gives the distribution of b_{i+1} which yields the same.

The monotonicity of $V_B^*(\cdot)$ then follows from Proposition 5 of Smith & McCardle (2002). □

C.8 PROOF OF THEOREM 5: OPTIMAL BIASED POLICY

We first prove that the optimal value function is continuous in the following lemma:

Lemma 16. Biased value function continuity

$V_B^*(\cdot)$ is continuous w.r.t. b .

Proof. First consider the one-stage value function defined as $V_B^0(b) = \sup_{q \in [0.5, 1]} r_B(b, q)$. From Theorem 4, we know that $\pi_B^{\epsilon, 0}(b) \in \{1 - b, 1, b - \epsilon, p\}$. Thus, we can equivalently write the one-stage value function as follows:

$$\begin{aligned} V_B^0(b) &= \sup_{\epsilon > 0} r_B(b, \pi_B^{\epsilon, 0}(b)) \\ &= \sup_{\epsilon > 0} \max_{q \in \{p, 1 - b, 1, b - \epsilon, p\}} r_B(b, q) \end{aligned} \quad (48)$$

Using the above expression, we now argue that $V_B^0(\cdot)$ is continuous on $b \in [0, 1]$. First, consider $b \leq 0.5$. From Theorem 4, an ϵ -optimal policy applies precision $1 - b$ or p for $b \leq 0.5$. In either case, the instantaneous reward will not depend upon ϵ , and the supremum in (48) can be omitted, leaving $V_B^0(b) = \max_{q \in \{1 - b, p\}} r_B(b, q)$. For $b \leq 0.5$, $r_B(b, 1 - b)$ falls in case one of (44) and is thus continuous with respect to b . On the same interval, $r_B(b, p)$ has one discontinuity at $b = 1 - p$ where it switches from case two to case one of (44). At $b = 1 - p$, Theorem 4 shows that the optimal myopic policy switches from $1 - b$ to p . Both policies fall in the first case of

(44), and, at $b = 1 - p$, $p = 1 - b$, so the controls are equal. Thus, the left and right limits coincide with the optimal value, and $V_B^0(\cdot)$ is continuous.

For $b \geq 0.5$, Theorem 4 gives the candidate ϵ -optimal policies $q \in \{1, b - \epsilon, p\}$. When the inner maximum in (48) is attained for $q = b - \epsilon$, the resulting value is $V_B^0(b) = \sup_{\epsilon > 0} r_B(b, b - \epsilon)$. From the third case of (44), we can rewrite this as $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|b - \epsilon - p|)$. Since $b - \epsilon$ is only selected for $b \leq p$ (Theorem 4), this reduces to $V_B^0(b) = -\beta(|p - b|)$. Thus, exchanging the order of the maximum and supremum in (48), we obtain $V_B^0(b) = \max\{r_B(b, 1), -\beta(|p - b|), r_B(b, p)\}$ for $b \geq 0.5$. For the first expression in this maximum, $r_B(b, 1)$ falls into case one of (44) and is thus continuous. The continuity of $-\beta(|p - b|)$ follows from the continuity of $\beta(\cdot)$. For $r_B(b, p)$, since $b \geq 0.5$, the first case of (44) applies for $b \leq p$ and the third case applies for $b > p$. Thus, $r_B(b, p)$ is potentially discontinuous at $b = p$. Therefore, what remains is to show the continuity of $V_B^0(\cdot)$ at $b = p$.

At $b = p$, the myopic ϵ -optimal policy switches from $q = b - \epsilon$ to $q = p$ by Theorem 4 with $\pi_B^{\epsilon, 0}(p) = b - \epsilon$. Thus, for $b = p$, the inner maximum of (48) is attained at $q = b - \epsilon$ and, from the third case of (44), $V_B^0(p) = \sup_{\epsilon > 0} -\beta(\epsilon) = 0$. Similarly, from Theorem 4, for each $\epsilon > 0$, there exists $t_\epsilon < p$ such that on $[t_\epsilon, p]$ the inner maximum of (48) is attained by $q = b - \epsilon$. On this interval, $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|p - b + \epsilon|) = -\beta(|p - b|)$. Therefore, $\lim_{b \uparrow p} V_B^0(b) = 0$. For $b > p$, the myopic optimal policy is $q = p$ from Theorem 4, and this yields reward $V_B^0(b) = 0$ for all $b > p$ from the third case of (44). This proves the continuity of $V_B^0(\cdot)$ at $b = p$ and completes the proof that $V_B^0(\cdot)$ is continuous on $[0, 1]$.

Applying Theorem 4.2 of Hinderer (2005), then yields the continuity of $V_B^*(\cdot)$. $\square$

Remark 4. Given the continuity of $V_B^*(\cdot)$ with respect to public belief and the fact that the control variable lies in the compact set $[0.5, 1]$, one might conclude that the optimal is guaranteed to exist at all public beliefs. However, there is an important subtlety which prevents such a conclusion: while $V_B^*(\cdot)$ is continuous with respect to public belief, it need not be continuous with respect to precision. Specifically, b_{i+1} is a function of both b_i and q_i (see Equation (3)); thus, $\mathbb{E}[V_B^*(b_{i+1})]$ may be denoted as $\phi(b_i, q_i)$. Now, b_{i+1} need not be continuous with respect to q_i , thus $\phi(\cdot)$ may not be continuous with respect to q_i . Thus, there may be regimes in which the optimal value is not attained by any policy. Indeed, we will show that such cases exist.

Proof of Theorem 5 We now leverage Lemma 16 to prove Theorem 5. We again let $z(b, q) = b + q - 2bq$ be the probability of an agent receiving signal B with precision q and prior public belief b and $y(b, q) = 1 - z(b, q)$ be the corresponding probability of the agent receiving signal G . We abuse notation to allow $V_B(b, q)$ to be the expected utility when beginning at public belief b , applying precision q in the current time step, and then following the optimal policy $\pi_B^*(\cdot)$ in all future steps. We can express $V_B(b, q)$ as follows by applying (13), (3), and (44):

$$V_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) \\ + \delta y(b, q) V_B^*(f(b, q, G)) \\ + \delta z(b, q) V_B^*(f(b, q, B)) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) \\ -C\mathbf{1}(b < 0.5) \\ + \delta V_B^*(b) & \text{otherwise} \end{cases} \quad (49)$$

Consider the latter case of (49) when $q < \max(b, 1 - b)$. Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by choosing q as close to p as possible. Thus, if the optimal precision is less than $\max(b, 1 - b)$, then it will be as close to p as possible.

We now consider four cases: (1) $b \in [0, 1 - p]$, (2) $b \in [1 - p, 0.5]$, (3) $b \in [0.5, p]$, and (4) $b \in (p, 1]$.

Case 1: $b \in [0, 1 - p]$, claims (A) and (B)

In this case, we will prove the existence of t_1 such that $\pi_B^*(b) = p$ for $b \leq t_1$. First note that if $\pi^*(b) < \max(b, 1 - b)$, we fall into the second case of Equation (49). Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by minimizing $|q - p|$. Because $p \in [0.5, 1]$, when $b \in [0, 1 - p]$, $p < \max(b, 1 - b)$. Thus, for $b \in [0, 1 - p]$, if $\pi_B^*(b) < \max(b, 1 - b)$, then $\pi_B^*(b) = p$.

Now, we evaluate the value function at belief $b = 0$. At $b = 0$, the state of the world is known to be B and, thus, no signal, regardless of its precision, will move the public belief. This can be seen mathematically by substituting $b = 0$ into the updated belief expressions in (13). Furthermore, at $b = 0$, if $q \geq \max(b, 1 - b)$, then $q = 1$. As argued above, if the precision is less than $\max(b, 1 - b)$, the optimal precision is $q = p$. We can compare the resultant value under these two candidate policies as follows from Equation (49):

$$V_B(0, 1) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (50)$$

$$V_B(0, p) = -C + \delta V_B^*(0) \quad (51)$$

Because $V_B(0, 1) < V_B(0, p)$, $\pi_B^*(0) = p$ and $V_B^*(0) = V_B(0, p)$. Thus, from Equation (51), $V_B^*(0) = \frac{-C}{1-\delta}$. By Lemma 15, this is a lower bound for the value function at any public belief. Therefore, and since $V_B^*(b) \leq 0$, $|V_B^*(b)| \leq \frac{C}{1-\delta}$ for all $b \in [0, 1]$.

We now show that there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). Thus, in this interval the optimum control lies in the second case of Equation (49) so $\pi_B^*(b) < \max(b, 1-b)$. It follows from our initial argument earlier in this case that $\pi_B^*(b) = p$ in this interval, which would complete the proof.

We define function $g(\cdot)$ to be the maximum of the first case of the RHS of Equation (49):

$$g(b) = \max_{q \geq \max(b, 1-b)} \left[-\beta(|q-p|) - Cz(b, q) + \delta y(b, q) V_B^*(f(b, q, G)) + \delta z(b, q) V_B^*(f(b, q, B)) \right] \quad (52)$$

For $b \in (0, 1)$, $f(b, q, G)$ and $f(b, q, B)$ are both continuous with respect to q from Equation (13). Thus, the maximum in Equation (52) must exist because it is of a function which is continuous with respect to q over a compact set $q \in [\max(b, 1-b), 1]$.

We now take the limit of $g(\epsilon)$ as ϵ tends to 0. Because $q \in [\max(\epsilon, 1-\epsilon), 1]$, when $\epsilon \rightarrow 0$, $q \rightarrow 1$ (inside the maximum in Equation (52)). We now reason about each of the four terms in Equation (52). As $\epsilon \rightarrow 0$, $q \rightarrow 1$ so that 1) from the continuity of $\beta(\cdot)$, $\beta(|q-p|) \rightarrow \beta(1-p)$, 2) from the continuity of $z(\cdot)$, $z(\epsilon, q) \rightarrow 1$, $y(\epsilon, q) \rightarrow 0$. Thus, the second term goes to C . Because $V_B^*(\cdot)$ is bounded and $y(\epsilon, q) \rightarrow 0$, the third term goes to 0. Finally, because $z(\epsilon, q) \rightarrow 1$ and $f(\epsilon, q, B) \rightarrow 0$ and since $V_B^*(\cdot)$ is continuous from Lemma 16, the last term approaches $\delta V_B^*(0)$. Thus,

$$\lim_{\epsilon \rightarrow 0} g(\epsilon) = -\beta(1-p) - C + \delta V_B^*(0) \quad (53)$$

Applying precision p , however, yields the following from the second case of Equation (49):

$$\lim_{\epsilon \rightarrow 0} V_B(\epsilon, p) = \lim_{\epsilon \rightarrow 0} [-\beta(p-p) - C + \delta V_B^*(\epsilon)] \quad (54)$$

$$= -C + \delta V_B^*(0) \quad (55)$$

The last equality follows from the continuity of $V_B^*(\cdot)$ from Lemma 16.

Note that the expression in Equation (53) is strictly less than that in Equation (55). Thus, there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). The result follows.

Case 2: $b \in [1-p, 0.5)$, Claim (C)

We now argue that for $b_i \in [1-p, 0.5)$, $\pi^*(b_i) \geq 1-b_i$, yielding claim (C). Consider a stationary policy $\pi(\cdot)$ such that $\pi(b_i) < 1-b_i$. Note that, from Equation (3), the public belief will remain unchanged under such a policy. Thus, the same control will be applied at every future time step, and, from Equation (2), every agent will select action B . Using the facts that $1-b_i \leq p$ in this belief range, $\mathbb{P}(a_i = B | b_i, \pi(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the expected total discounted utility under such a policy as follows:

$$V_B^\pi(b_i) = -\beta(p - \pi(b_i)) - C + \delta V_B^\pi(b_i)$$

Thus, if this policy is applied, it will result in expected value $V_B^\pi(b_i) = \frac{-\beta(p - \pi(b_i)) - C}{1-\delta}$. That is, the cost of every agent choosing action B and continuing to apply precision $\pi(b_i)$. This is strictly lower than the value function lower bound of $\frac{-C}{1-\delta}$ established using Lemma 15 in Case 1 because $\pi(b_i) < 1-b_i \leq p$. Therefore, if an optimal policy exists at this point, it must be the case that $\pi_B^*(b_i) \geq 1-b_i$.

The remaining candidate policies are $q_i \in [1-b_i, 1]$. Applying Bellman's principle we can write the following:

$$V_B^*(b_i) = \sup_{q_i \in [1-b_i, 1]} r_B(b_i, q_i) + \delta \mathbb{E}[V_B^*(b_{i+1})] \quad (56)$$

Because $q_i \geq \max(b_i, 1-b_i)$ and $b \in [1-p, 0.5) \subset (0, 1)$, Equation (13) and Equation (3) imply that $b_{i+1} = \frac{b_i q_i}{y(b_i, q_i)}$ or $b_{i+1} = \frac{b_i(1-q_i)}{z(b_i, q_i)}$. Both of these expressions are continuous with respect to q_i and $V_B^*(\cdot)$ is continuous by Lemma 16, thus the RHS of Equation (56) inside the supremum is continuous with respect to q_i . Therefore, in Equation (56), the supremum over a compact set $q_i \in [1-b_i, 1]$ of a function which is continuous with respect to q_i must be attained, implying the existence of an optimal policy. Thus, $\pi_B^*(b_i) \geq 1-b_i$.

Case 3: $b \in [0.5, p]$ Claims (D) and (E)

For $b = 0.5$, $q \geq b$ as we assumed $q \in [0.5, 1]$. This places us in the first case of Equation (49). Furthermore, $\beta(\cdot)$, $y(b, q)$, $z(b, q)$, $f(b, q, G)$, and $f(b, q, B)$ are all continuous with respect to q for $b = 0.5$. Thus, the

supremum over $q \in [0.5, 1]$ of $V(b, q)$ must be attained as it is of a continuous function of q over a compact set of q . Therefore, an optimal policy must exist at $b = 0.5$.

Now consider $b \in (0.5, p]$ and a stationary policy $\pi'(\cdot)$ such that $\pi'(b) < b$. From Equation (3) and Equation (2), applying such a policy will result in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. Using the facts that $b_i \leq p$, $\mathbb{P}(a_i = G | b_i, \pi'(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the total expected discounted utility under such a policy as follows:

$$V_B^{\pi'}(b_i) = -\beta(p - \pi'(b_i)) + \delta V_B^{\pi'}(b_i)$$

Thus, the policy results in expected value $V_B^{\pi'}(b_i) = \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$. Note that, in this case, $V_B^{\pi'}(b_i)$ is an increasing function of $\pi'(b_i)$. Thus, the supremum of $V^*(\pi')_B(b_i)$ over $\pi'(b_i) \in (0.5, b)$ is not attained.

We now consider two cases: (1) $\max_{q_i \in [b_i, 1]} V(b_i, q_i) \geq \sup_{\pi'(b_i) \in (0.5, b_i)} \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$, and (2) the complement. In the former case, we can argue as we did for $b = 0.5$ that the supremum must be attained and an optimal policy exists such that $\pi_B^*(b_i) \geq b_i$. In the latter case, as argued above, no optimal policy exists. Instead, for sufficiently small $\epsilon > 0$, there exists an ϵ -optimal policy such that $\pi_B^\epsilon(b_i) = b_i - \epsilon$. This yields claims (D) and (E).

What remains is to show that there exists $t_2 \in (0.5, p)$ such that for $b_i \in (t_2, p]$ we fall into the latter case, yielding claim (F). We define functions $m(\cdot)$ and $n(\cdot)$ as follows corresponding to the resultant value from choosing precisions less than b or greater than or equal to b , respectively:

$$\begin{aligned} m(b) &= \sup_{\pi'(b) \in [0.5, b)} \frac{-\beta(p - \pi'(b))}{1 - \delta} \\ n(b) &= \max_{q \geq b} V(b, q) \end{aligned}$$

Taking the limit of $m(\cdot)$ as $b \rightarrow p$ yields:

$$\lim_{b \rightarrow p} m(b) = 0 \tag{57}$$

When $q \geq b$ we fall into the first case of Equation (49). Thus we can expand $n(b)$ as follows:

$$\begin{aligned} n(b) &= \max_{q \geq b} [-\beta(|q - p|) - Cz(b, q) \\ &\quad + \delta y(b, q)V_B^*(b, q) \\ &\quad + \delta z(b, q)V_B^*(b, q)] \end{aligned} \tag{58}$$

We will argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Since p is a fixed parameter strictly less than 1, this means from Equation (57) that $n(b) \leq -C(1 - p) < \lim_{b \rightarrow p} m(b)$, which implies the existence of an open interval $(t_2, p]$ where an optimal policy does not exist, yielding claim (F). We now argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Note that every term of Equation (58) is bounded above by 0, so it suffices to show that any one term, say the second term $-Cz(b, q)$, is upper bounded by $-C(1 - p)$. First note that $z(b, q) = q(1 - 2b) + b$ so $z(b, q)$ is decreasing in q since $b > 0.5$. Thus, $z(b, q) \geq z(b, 1) = 1 - b \geq 1 - p$. The last inequality follows because $b \in (0.5, p]$. The claim follows since $p < 1$.

Along with Equation (57), this implies the existence of an open interval (t_2, p) where an optimal policy does not exist, yielding claim (F).

Case 4: $b \in (p, 1]$ Claim (A)

For $b \in (p, 1]$, consider applying stationary policy $\pi(\cdot)$ such that $\pi(b) = p$. Because $b > p$, Equation (3) and Equation (2) imply that such a policy results in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. We can apply Equation (6) to write

$$V_B^\pi(b) = -\beta(p - p) + \delta V_B^\pi(b) = \delta V_B^\pi(b)$$

Thus, $V_B^\pi(b) = 0$. Since this is also an upper bound of the optimal value function for any b , $\pi_B^*(b) = p$. ■

C.9 SOCIAL WELFARE MONOTONICITY

Lemma 17. *Precision Monotonicity of Social Welfare*

If $\pi_1(b) \leq \pi_2(b) \forall b \in [0, 1]$, for policies $\pi_1(\cdot), \pi_2(\cdot)$, then

$$W^{\pi_1}(b) \leq W^{\pi_2}(b) \forall b \in [0, 1]$$

Proof. It is clear that the one-step welfare $-C \min(b_i, 1 - b_i, \pi(b_i))$ is increasing in $\pi(b_i)$.

Furthermore, note that the convexity of social welfare with respect to public belief follows from the same argument as in the proof of Theorem 2. This is because social welfare is equivalent to the altruistic value function in the special case where $\beta(\cdot) = 0$.

We can then follow the same argument as in the proof of Lemma 9 (Appendix C.4) to show that increasing precision increases welfare. $\square$

D EXTENSION TO HETEROGENEOUS AGENT PREFERENCES

We show that, for the altruistic planner, the model is equivalent to a setting with two observable agent types that have opposing preferences.

Assume that there are two agent types, G and B , with the type of agent i denoted t_i . The first type receives unit reward when $a_i = \omega$ and no reward otherwise. The second receives unit reward when $a_i \neq \omega$ and no reward otherwise. Further, assume that all agents know their own type and the types of all other agents. All other aspects of the model remain unchanged.

Agent i 's private belief update is unchanged because it depends only on the signal structure, i.e., $\mathbb{P}(s_i = \omega) = q_i$:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q}{1+2b_i q_i - b_i - q_i} b_i & s_i = G \\ \frac{1-q}{b_i + q_i - 2b_i q_i} b_i & s_i = B \end{cases} \quad (59)$$

Using Equation (59), agent i 's action can be written as follows, where the superscript indicates the agent type and $!$ denotes negation:

$$a_i^G = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & b_i < 1 - q_i \end{cases}$$

$$a_i^B = \begin{cases} !s_i & 1 - q_i \leq b_i \leq q_i \\ B & q_i < b_i \\ G & b_i < 1 - q_i \end{cases}$$

Because each agent's type is publicly known, when $q_i \geq \max(b_i, 1 - b_i)$ all agents can perfectly infer agent i 's private signal from her action. If $q_i < \max(b_i, 1 - b_i)$, agent i 's action is uninformative.

After observing agent i 's type t_i , precision q_i , and action a_i , the public belief updates as follows:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (60)$$

In this setting, an altruistic planner seeks to induce type- G agents to take action ω and type- B agents to take action $!\omega$, because these choices align with the agents' personal utility functions.

Thus, the planner's instantaneous reward can be expressed as follows, with the superscript denoting the type of agent i :

$$r^G(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i \neq \omega)$$

$$= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}$$

$$r^B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = \omega)$$

$$= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}$$

These expressions are identical and equal to the reward function in Equation (5). The altruistic planner's problem is therefore equivalent under heterogeneous but observable types. $\blacksquare$

E EXPERIMENTS

E.1 EXPERIMENTAL SETUP

To simulate controlled social learning, we consider a sequence of agents deciding whether to buy a new car model after observing past actions and an advertisement whose precision is selected by the planner. LLMs serve three distinct roles: (1) agents, (2) planner, and (3) oracle. Each agent is assigned a profile containing demographic information, car use cases, and preferences. At the outset, we select either the good or bad car as the offering. The information sets and objectives of the planner and agents match the model of Section 3.

The oracle has two roles: (1) it generates a generic message that serves as the baseline-precision signal without planner intervention, and (2) after the planner chooses a precision, it generates a message with the prescribed precision customized to the target agent’s profile.

The oracle receives a fact sheet for each type of car (safety rating, customer satisfaction, and related attributes) along with the agents’ profiles. Together, the fact sheet and profile support generation of a tailored message with the precision prescribed by the planner. Appendix E.2 provides additional detail on the experimental setup and prompting.

Each agent makes a binary decision based on observed previous decisions and her private signal. The agent’s objective is to match the true parameter value.

The planner receives the same decision history and chooses a precision level for each agent’s message. The planner is also given the precision of a generic, untailored message that can be sent at zero cost. In the altruistic case, the planner’s objective is to induce correct decisions; in the biased case, the planner’s objective is to induce agents to buy the car.

We compare the LLM simulations to the analytic solutions obtained via JuliaPOMDP (Egorov et al., 2017).

E.2 PARAMETERS AND PROMPTS

E.2.1 INSTANCE PARAMETERS

Experiments were run using the LG AI EXAONE Deep 32B model (LG AI Research, 2025). We solve the model analytically and run the LLM simulations using the parameters in Table 1.

Parameter	Description	Values
C	Cost of a bad action	1
k	Scaling factor for control cost	$\{0.1, 0.3, 0.5, 0.7, 0.9\}$
p	Baseline signal precision	$\{0.6, 0.7, 0.8, 0.9\}$
δ	Discount Factor	$\{0, 0.25, 0.5, 0.75, 1\}$

Table 1: Instance parameters

The car facts were designed so that the good car was good for everyone and the bad car was bad for everyone. Below we include the fact sheets for the GOOD and BAD cars, respectively.

```
fact_sheet["GOOD"] = [  
    "EPA-rated range: 290 miles.",  
    "Fast-charge 10-80 %% in 24 min on 150 kW DC.",  
    "8-year / 100,000-mile battery warranty.",  
    "NHTSA 5-Star overall safety rating.",  
    "Cold-soak -5 °F keeps 93 %% range.",  
    "Cools cabin 100→72 °F in 3 min.",  
    "Cargo volume 26 ft3 total.",  
    "Tow's 1 500 lb with factory hitch.",  
    "Average 1.2 service visits first 3 yrs.",  
    "92 %% owners would buy again (Apr-25).",  
    "Qualifies for federal credit.",  
    "Quarterly OTA updates incl. battery pre-heat."  
]  
fact_sheet["BAD"] = [  
    "EPA-rated range: 155 mi (combined).",  
    "Fast-charge 10 %% → 80 %% in 51 min (50 kW DC hardware limit).",  
    "Battery warranty: 3 yr / 36 000 mi, first owner only.",
```

```

1836         "NHTSA overall safety rating 3 Stars (frontal-offset weakness).",
1837         "Cold-soak 15 °F retains 58 %% range;cabin heat cuts range 20%.",
1838         "Cabin cools 105 °F → 72 °F in 11 min.",
1839         "Cargo volume 17 ft³; rear seats fixed (no fold-flat).",
1840         "No factory towing approval.",
1841         "Average 3.8 service visits per year (battery-alert issues).",
1842         'Owner survey: 41 %% "would buy again" (1 200 owners, Feb-25).',
1843         "Not eligible for federal tax credit.",
1844         "OTA updates limited to infotainment."
1845     ]

```

Profiles were generated programmatically. Profiles included an agent's name, age, city of residence, interests, planned use case for the car (e.g., commuting, road trips, etc.), and major pain points or concerns (e.g., fuel efficiency, winter driving, etc.).

E.2.2 PROMPTS

Agent LLM Prompt:

```

1852 sys = (
1853     f"You are deciding whether to buy a car. "
1854     f"Your profile is {profile}"
1855     f"You start believing there's a 50% chance the car is GOOD. "
1856     f"You have two information sources: "
1857     f"1. Previous agents' decisions "
1858     f"2. An advisor's message about the car "
1859     f"Your goal is to maximize your reward "
1860     f"i.e. BUY if the car is GOOD and "
1861     f"NOT BUY if the car is BAD. "
1862     f"If you BUY a GOOD car, you GAIN ${C}. "
1863     f"If you NOT BUY a BAD car, you GAIN ${C}. "
1864     f"If you BUY a BAD car, you LOSE ${C}. "
1865     f"If you NOT BUY a GOOD car, you LOSE ${C}. "
1866     f"belief = your probability (0.0-1.0) that the car is GOOD. "
1867     f"decision = exactly 'BUY' or 'NOT BUY'."
1868 )
1869 user = (
1870     f"PREVIOUS DECISIONS:\n{history_txt or '(none)'}\n\n"
1871     f"ADVISOR MESSAGE:\n{message}\n\n"
1872     f"TASK: Use the previous decisions and the advisor message. "
1873     f"Decide BUY or NOT BUY. "
1874     f"Update your belief about car quality (0.0-1.0). "\n"
1875     f'JSON response:{{"belief": <number>,"decision": "<BUY or NOT BUY>"}}'
1876 )

```

Planner LLM Prompt:

```

1876 sys = (
1877     f"You are providing messages to potential car buyers. "
1878     f"Each potential car buyer (or agent) has the same information. "
1879     f"Your goal is to get agents to {objective}."
1880     f"Each decision according to your objective, earns you {C} dollars. "
1881     f"You and the agents have the same information. "
1882     f"Initially there is a 50% chance the car is GOOD. "
1883     f"The agents' aim to BUY if the car is GOOD "
1884     f"and NOT BUY if the car is BAD. "
1885     f"You can send a baseline message for free "
1886     f"or tailor the message to the agent at a cost. "
1887     f"Tailoring costs {k} dollars times the change in precision. "
1888     f"(Message precision is the probability (0.5 to 1.0) "
1889     f"that the message leaves the buyer with the correct impression. "
1890     f"For example, if the car is likely GOOD: "
1891     f"a high precision message will likely suggest BUY "
1892     f"and a low precision message will likely suggest NOT BUY. "
1893     f"You may choose to increase or decrease precision. "

```

```

1890     f"Agents decide based on your message and the decision history. "
1891     f"They update beliefs by observing previous decisions. "
1892     f"IMPORTANT: Thus your objective is to maximize:"
1893     f"{C} * (# of agents who choose according to your objective) "
1894     f"- sum of ({d}^i*{k}*abs(message_precision - baseline_precision))\n"
1895     f"where {d} is your discount factor for future cost. "
1896 )
1897 user = (
1898     f"BASELINE MESSAGE precision:\n{pi_base:.2f}\n"
1899     f"DECISION HISTORY:\n{history_txt}\n"
1900     f"TASK:"
1901     f"Based on the history and your objective, "
1902     f"choose your desired precision.\n"
1903     f'JSON response: {"message_precision": <number>}}\n'
1904 )

```

Oracle LLM Prompt:

```

1905 sys = (
1906     f"You will be given "
1907     f"1. an agent profile, "
1908     f"2. information about a car, "
1909     f"3. the true quality of the car, "
1910     f"4. a baseline message and the 'precision' of that message, and "
1911     f"5. a desired precision for the message you output. "
1912     f"(Message precision is the probability (0.5 to 1.0) "
1913     f"that the message leaves the buyer with the correct impression. "
1914     f"For example, if the car is likely GOOD: "
1915     f"a high precision message will likely suggest BUY "
1916     f"and a low precision message will likely suggest NOT BUY. "
1917     f"IMPORTANT: Output a message with the requested precision. "
1918     f"IMPORTANT: The message should be tailored to the profile. "
1919 )
1920 user = (
1921     f"AGENT PROFILE:{profile}\n"
1922     f"TRUE CAR QUALITY: {omega}\n\n"
1923     f"CAR FACTS: {car_facts}\n"
1924     f"BASELINE MESSAGE:{baseline_message}}\n"
1925     f"BASELINE MESSAGE precision: {p}\n"
1926     f"DESIRED MESSAGE precision: {q}\n"
1927     f"IMPORTANT: Output a message with the requested precision. "
1928     f"TASK: Write a message (MAX 100 words) with the desired precision."
1929     f'JSON response: {"message": "<message>}}'
1930 )

```

E.3 VALIDATION OF LLM BELIEFS AND ORACLE

We provide additional experiments validating the self-reported beliefs of the LLMs (as shown in figure 1b) and the performance of the LLM oracle. In these experiments, we treat the LLMs as rational and utility-maximizing, but not necessarily Bayesian, agents. The goals are (1) to show that the model's self-reported posterior corresponds to the belief governing its choices in state-contingent decision problems, and (2) to verify that messages produced by the oracle have the requested precision.

First, we provide the LLM with a prior and a new signal, obtain its self-reported posterior belief, and then ask it to make a binary “buy/not-buy” decision that yields unit reward if and only if the true state is good. A rational agent should choose “buy” exactly when its subjective belief exceeds 0.5. As shown in figure 4a, the LLM's decisions align sharply with this threshold: it overwhelmingly chooses “buy” when its reported posterior exceeds 0.5 and “not buy” otherwise. Next, we elicit the model's willingness to pay for a payoff contingent on the true state. For a range of prices c , we ask whether the agent would pay c in exchange for receiving unit utility if the state is good. A rational agent with belief p should accept exactly when $c < p$. Figure 4b plots the fraction of “buy” decisions across posterior–price pairs and shows a clear separation: acceptance rates are high when $c < p$ and low when $c > p$, as expected from a utility-maximizing decision maker whose belief is p . Both experiments indicate that the self-reported posterior is close to the belief the model uses in decision making.

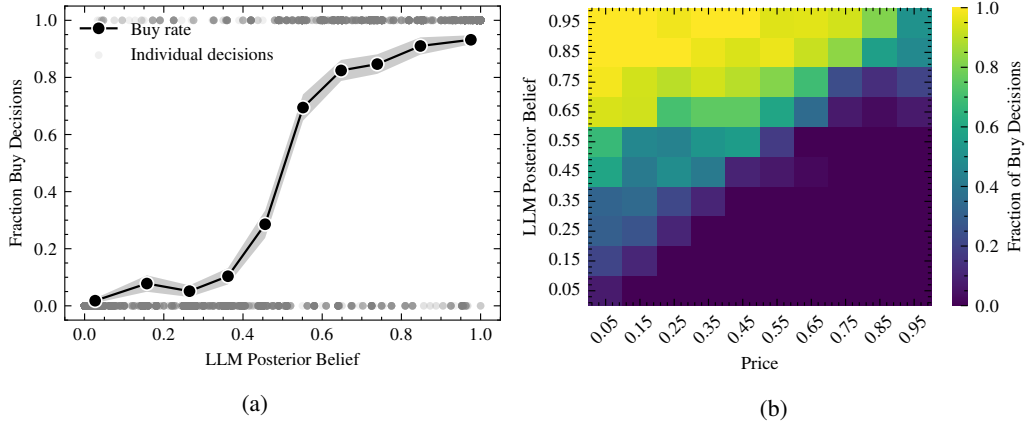


Figure 4: Rate at which the LLM chooses a ‘buy’ decision as a function of (a) its self-reported posterior belief and (b) its self-reported posterior together with the purchase price.

We also validate the oracle’s ability to craft messages of a specified precision. After the oracle generates a message with requested precision q , we ask another LLM to estimate the message precision $\hat{q}$. We conduct this procedure for $q \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ on 100 programmatically generated agent profiles with 10 trials for each combination, for a total of 6,000 trials. The mean absolute error $|q - \hat{q}|$ was 0.025, indicating that realized message precision closely follows the requested precision.

E.4 IMPACT OF NON-BAYESIAN LLM BELIEF UPDATES ON LEARNING TRAJECTORIES

We unpack belief trajectories from an example LLM experiment to show how non-Bayesian belief updating affects learning dynamics. We fix the policy to be the analytically optimal policy and apply it to LLM-simulated agents. When the car is bad (right panel), the altruistic planner drives the Bayesian belief (dotted black line) rapidly toward 0. The LLM belief (solid black line), however, is more persistent. This aggregate trajectory follows from the micro-level biases: the public belief resists entering a cascade (NB3) because individual agents overreact to occasional positive signals that run counter to the group’s increasingly negative belief (NB2).

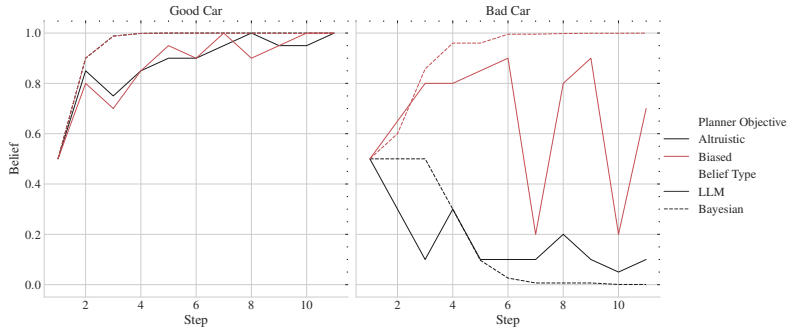


Figure 5: Agent belief-updating trajectories for the true Bayesian belief and the belief reported by the LLM agent under analytically optimal policies for different planner objectives and true states.